Diretrizes de Terapia Intensiva

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MCG Health -
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Esta diretriz de cuidados da MCG ainda não foi atualizada para refletir as novas orientações e requisitos encontrados na Regra Final do Medicare Advantage e da Parte D de 2024 (CMS-4201-F). Para beneficiários do Medicare Advantage, antes de consultar as diretrizes de cuidados da MCG, os usuários devem primeiro seguir todas as orientações relacionadas a este tópico encontradas nas Determinações de Cobertura Nacional (NCDs) e nas Determinações de Cobertura Local (LCDs) do CMS, bem como em outros estatutos e regulamentos. O conteúdo utilizado nas diretrizes de cuidados da MCG pode ser usado para complementar, mas não substitui, modifica ou revoga os regulamentos existentes do Medicare ou as NCDs ou LCDs aplicáveis.

- Diretrizes de Admissão em Unidade de Terapia Intensiva
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- Referências
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Diretrizes de Admissão em Unidade de Terapia Intensiva

- A admissão na UTI pode ser indicada quando a necessidade for demonstrada por **um ou mais** dos seguintes critérios :(1)(2)(3)(4)(5)(6)(7)(8)(9)(10)(11)(12):
  - Anormalidades nos sinais vitais, incluindo **1 ou mais** dos seguintes [A](5)(11)(13)(14)(15):
    - Pressão arterial sistólica inferior a 90 mmHg, ou 20 mmHg abaixo da pressão arterial habitual do paciente em adultos ou crianças com mais de 10 anos.(14)(16)
    - Pressão arterial sistólica inferior a 60 mmHg em recém-nascidos (menos de 1 mês de idade), 70 mmHg em lactentes (de 1 mês a 1 ano de idade) ou inferior à soma de 70 mmHg mais o dobro da idade do paciente em crianças de 2 a 10 anos de idade.(14)
    - Pressão arterial média inferior a 65 mmHg em adultos ou crianças com 10 anos ou mais [B] Calculadora de PAM(5)(9)
    - Queda da pressão arterial média [B] superior a 30 mmHg em relação ao valor basal, atribuível à disfunção cardiovascular (por exemplo, choque cardiogênico). Calculadora de PAM(15)
    - Pulso inferior a 45 ou superior a 130 batimentos por minuto em adultos.(17)
    - Pulso inferior a 40 ou superior a 140 batimentos por minuto em adolescentes com 16 anos ou mais.
    - Pulso inferior a 70 ou superior a 150 batimentos por minuto em crianças de 2 a 15 anos de idade.
    - Pulso inferior a 90 ou superior a 160 batimentos por minuto em lactentes com menos de 2 anos de idade.
  - Resultados laboratoriais (novos), incluindo **um ou mais** dos seguintes:(18)(19)(20)(21)(22)(23)(24)(25)(26):
    - Oximetria de pulso ou saturação arterial de oxigênio inferior a 90%, ou pressão parcial de oxigênio inferior a 60 mm Hg (8,0 kPa) apesar da suplementação de oxigênio [C](17)
    - Aumento da pressão parcial de dióxido de carbono com acidose respiratória aguda ou não compensada
    - pH arterial inferior a 7,2 ou superior a 7,65
    - Sódio sérico inferior a 110 mEq/L (mmol/L) ou superior a 160 mEq/L (mmol/L)(24)
    - Níveis séricos de sódio inferiores a 125 mEq/L (mmol/L) associados a alteração do estado mental , convulsão ou parada respiratória.(24)

- Níveis séricos de sódio superiores a 145 mEq/L (mmol/L) associados a alteração do estado mental , convulsões, diabetes insípido que requerem monitoramento rigoroso do equilíbrio hídrico e ressuscitação, ou desidratação grave que exige reposição volêmica com grandes volumes de fluidos.
- Potássio sérico inferior a 2 mEq/L (mmol/L) ou superior a 7 mEq/L (mmol/L)
- Potássio sérico inferior a 2,5 mEq/L (mmol/L) com manifestações clínicas ou eletrocardiográficas graves (por exemplo, paralisia, arritmia cardíaca, despolarização tardia, ondas T planas ou ondas U).
- Potássio sérico superior a 6,5 mEq/L (mmol/L) com alterações eletrocardiográficas de hipercalemia (por exemplo, ondas T apiculadas, QRS alargado, arritmia cardíaca; para pacientes com marca-passo permanente: QRS estimulado alargado, aumento do intervalo entre o estímulo do marca-passo e o QRS ou falha na captura).(29)
- Cálcio [D] superior a 14 mg/dL (3,5 mmol/L) ou cálcio ionizado superior a 10,0 mg/dL (2,50 mmol/L)(32)
- Cálcio [D] superior a 12 mg/dL (3,0 mmol/L) ou cálcio ionizado superior a 8,0 mg/dL (2,0 mmol/L) com alteração do estado mental , depleção grave de volume que requer ressuscitação volêmica intravenosa com grandes volumes, hipotensão , arritmia significativa, bloqueio cardíaco ou toxicidade digitálica.(32)
- Fósforo sérico inferior a 1 mg/dL (0,32 mmol/L)(23)
- Nível tóxico da droga ou intoxicação que cause ou possa causar anormalidades neurológicas ou instabilidade hemodinâmica.
- Anormalidades laboratoriais menos graves que contribuem para **uma ou mais** das seguintes condições:
  - Convulsão
  - Alteração do estado mental
  - Fraqueza muscular ou espasmos severos
  - Arritmias
  - Instabilidade hemodinâmica
  - Outras manifestações clínicas significativas

- Resultados do eletrocardiograma (ou monitoramento cardíaco), incluindo **um ou mais** dos seguintes:
  - Arritmia inerentemente instável ou com risco de vida (ex.: taquicardia ventricular sustentada, fibrilação ventricular, assistolia)
  - Arritmia causando hipotensão (ex.: bradicardia , taquicardia)
  - Bloqueio cardíaco completo causando hipotensão.
  - Outros achados que indicam a necessidade de cuidados intensivos (por exemplo, isquemia cardíaca aguda, infarto do miocárdio)
- Achados físicos, incluindo **um ou mais** dos seguintes:
  - Vias aéreas ameaçadas
  - Alteração do estado mental grave ou persistente
  - Convulsões repetidas ou prolongadas
  - Anúria de início recente (débito urinário inferior a 0,1 mL/kg/hora durante 4 horas)
  - Cianose (nova)
  - Tamponamento cardíaco
  - Estado pós-parada respiratória ou cardíaca
  - Achados compatíveis com emergência abdominal (ex.: sinais de irritação peritoneal)
- Achados de imagem, como dissecção de aneurisma ou ruptura de víscera.
- Queimaduras graves, conforme indicado por **um ou mais** dos seguintes [E](33)(34)(35)(36)(37)(38)(39):
  - Instabilidade hemodinâmica
  - Hipotensão ou necessidade de reposição volêmica agressiva
  - Lesão por inalação(33)(40)
  - Insuficiência respiratória com necessidade de oxigênio de alto fluxo, ventilação mecânica ou oxigenação por membrana extracorpórea.(33)(40)
  - Necessidade de intubação ou ventilação mecânica devido ao alto risco de obstrução das vias aéreas (por exemplo, lesão laríngea observada na laringoscopia com fibra óptica, queimadura orofaríngea, fuligem nas narinas ou orofaringe, queimadura circular profunda no pescoço, estridor, alteração na voz ou queimadura facial extensa envolvendo uma área de superfície corporal total de 40% ou mais em adultos ou 60% ou mais em crianças).(33)(40)
  - Necessidade de intubação ou ventilação mecânica devido à redução do nível de consciência (ex.: escala de coma de Glasgow inferior a 8) Calculadora de ECG(33)
  - Necessidade de desbridamento e curativos frequentes ou intensivos.

- Queimadura química grave(41)(42)
- Queimadura elétrica de alta voltagem (por exemplo, 1000 volts ou mais)(34)(43)
- Intoxicação por monóxido de carbono
- Toxicidade do cianeto
- Bebê com menos de 1 ano de idade
- Múltiplas comorbidades
- Infecção grave ou sepse (ex.: pele e tecidos moles, pneumonia, infecção urinária)
- Disfunção cardíaca, renal, pulmonar ou neurológica com risco de vida
- Trauma concomitante, falência de órgãos, arritmia ou outra condição médica que exija cuidados intensivos.

− Intervenção ou monitoramento específico necessário, conforme indicado por **um ou mais** dos seguintes itens:(9)(10)(16):

- Nova necessidade de ventilação assistida, invasiva ou não invasiva [F](10)(44)(45)(46)
- Nova necessidade de intubação (por exemplo, para proteger as vias aéreas)
- Traqueostomia recente (menos de 48 horas)
- Verificação dos sinais vitais ou exames neurológicos a cada hora.
- É necessário monitoramento da linha da artéria pulmonar.
- É necessário monitoramento contínuo da linha arterial.
- Medicamentos vasoativos intravenosos contínuos
- Antiarrítmicos intravenosos contínuos
- Ressuscitação com grandes volumes de fluidos intravenosos (por exemplo, mais de 6 litros por dia)
- Necessidades transfusionais extensas ou rápidas (por exemplo, mais de 6 unidades em 24 horas)
- Tratamento intravenoso de alto risco, como trombólise, solução salina hipertônica ou infusão de manitol.
- Dessensibilização rápida ou teste de provocação medicamentosa gradual para reações de hipersensibilidade de alto risco à medicação necessária (ex.: penicilina, aspirina, anticorpo monoclonal).(47)(48)(49)
- Estimulação cardíaca aguda
- Bomba de balão intra-aórtico(50)(51)(52)(53)
- Dispositivo de assistência ventricular(50)(51)(52)(53)
- Dispositivo de oxigenação por membrana extracorpórea(50)(52)(54)
- Pericardiocentese
- Hemodiálise em paciente instável
- Terapia contínua de substituição renal (ex.: hemodiálise venovenosa contínua)(55)(56)(57)(58)
- Remoção contínua de fluidos por meio de hemofiltração (ex.: hemofiltração venovenosa contínua)
- Início da diálise peritoneal(59)
- Terapia broncoscópica de emergência (ex.: para hemoptise)
- Terapia endoscópica de emergência para sangramento
- Tamponamento com balão para sangramento varicoso
- Monitoramento da pressão intracraniana ou monitoramento da oxigenação tecidual(60)(61)
- Monitoramento de ventriculostomia
- Tratamento de crises convulsivas contínuas
- Hipotermia ou coma induzidos
- Testes e tratamentos frequentes e contínuos para condições agudas, incluindo **uma ou mais** das seguintes:
    - Correção de acidose ou alcalose metabólica grave
    - Verificações frequentes de glicose (ou seja, mais frequentes do que as que seriam possíveis em um nível de atendimento inferior).
    - Sobrecarga grave de fluidos
    - Edema cerebral
    - Monitoramento ou aspiração de secreções em casos de insuficiência respiratória ou acidose.

        - Monitorar sangramento ativo
    - Outras necessidades de tratamento ou monitoramento não disponíveis fora da UTI

- **Condições sistêmicas** , incluindo **uma ou mais** das seguintes:
    - Distúrbio eletrolítico ou metabólico grave que causa ou pode causar **um ou mais** dos seguintes sintomas:(18)(22)(23)(24)(25)(26)(29):
        - Disritmia cardíaca com risco de vida
        - Insuficiência respiratória
        - Alteração do estado mental
        - Convulsões
        - Instabilidade hemodinâmica
        - Fraqueza muscular(62)
    - Hipotermia grave , conforme indicado por **um ou mais** dos seguintes sintomas:(63)(64):
        - Temperatura de 32 graus Celsius (89,6 graus Fahrenheit) ou inferior.
        - Temperatura igual ou inferior a 35 graus Celsius (95 graus Fahrenheit) e disfunção orgânica grave, conforme indicado por **um ou mais** dos seguintes sintomas:
            - Alteração do estado mental
            - Instabilidade hemodinâmica
            - Depressão respiratória
            - Edema pulmonar
            - Arritmia (ex.: bradicardia , fibrilação atrial, fibrilação ventricular)
            - Lesão renal aguda (estágio 2) ou insuficiência renal aguda (estágio 3 da lesão renal aguda)
            - Outras novas disfunções de órgãos-alvo atribuíveis à hipotermia
    - Lesões ambientais como hipertermia, lesões elétricas ou quase afogamento.(63)(65)(66)(67)
    - Anafilaxia com comprometimento respiratório, hipotensão ou disfunção de outros órgãos-alvo.(68)(69)(70)
    - Hipertermia maligna confirmada ou suspeita, comprovada por exposição a agente anestésico volátil ou succinilcolina e **um ou mais** dos seguintes critérios:(63)(71)(72):
        - Hipermetabolismo, evidenciado por aumento inadequado da produção de CO_2 , consumo de O_2 ou acidose metabólica e respiratória mista aguda.
        - Taquicardia inexplicável
        - Taquiarritmias cardíacas, extrassístoles ventriculares ou bigeminismo ventricular.
        - Espasmo do masseter
        - Rigidez muscular
        - Aumento rápido da temperatura corporal central
        - Aumento agudo nos níveis séricos de potássio, creatina quinase ou mioglobina.
    - Síndrome neuroléptica maligna, evidenciada pela exposição a um fármaco neuroléptico (p. ex., fenotiazina, butirofenona) ou a um fármaco depletor de dopamina (p. ex., alfa-metiltirosina), ou pela suspensão de um agente dopaminérgico (p. ex., levodopa, amantadina) e **um ou mais** dos seguintes:(63)(73)(74):
        - Rigidez muscular
        - Hipertermia
        - Alteração do estado mental
        - Instabilidade hemodinâmica
        - Aumento agudo da creatina quinase
    - Síndrome serotoninérgica (toxicidade serotoninérgica) evidenciada pela exposição a medicamentos que aumentam os níveis de serotonina no SNC (por exemplo, alguns antidepressivos, incluindo inibidores da recaptação de serotonina, inibidores da monoamina oxidase, opioides, estimulantes, triptanos, anticonvulsivantes, linezolida) e **um ou mais** dos seguintes:(74)(75):
        - Achados neurológicos significativos (ex.: agitação, hipomania, alucinações, alteração do estado mental , convulsões)
        - Hiperatividade significativa relacionada ao sistema nervoso autônomo (ex.: instabilidade hemodinâmica , hipertensão, sudorese, hipertermia, taquicardia, vômitos, diarreia)
        - Achados musculoesqueléticos significativos (ex.: rigidez, mioclonia, hiperreflexia, tremor, rabdomiólise, elevação da creatina quinase)
    - Síndrome de abstinência alcoólica grave com **um ou mais** dos seguintes sintomas:(76)(77)(78):

- Instabilidade hemodinâmica
- Arritmias cardíacas de preocupação imediata
- convulsões incontroláveis
- Depressão respiratória com necessidade ou alta probabilidade de necessidade de ventilação mecânica.
- Necessidade de um agente anestésico para controlar a agitação (ex.: dexmedetomidina, propofol, cetamina, fenobarbital intravenoso)
- Delirium tremens, evidenciado por **TODOS** os seguintes sintomas:
  - Cessação ou redução do consumo excessivo e prolongado de álcool.
  - Delírio (ex.: distúrbio flutuante da atenção, consciência, orientação, linguagem)
  - Sinais ou sintomas de abstinência alcoólica, conforme indicado por **dois ou mais** dos seguintes:
    - Hiperatividade autonômica (diaforese, taquicardia, hipertensão)
    - Tremor
    - Náuseas ou vômitos
    - Alucinações
    - Aumento da ansiedade
    - agitação psicomotora
    - Crises tônico-clônicas generalizadas

- Diagnósticos ou procedimentos **cardiológicos , incluindo um ou mais** dos seguintes:(79)(80)(81)(82):
  - Dor no peito acompanhada de **um ou mais** dos seguintes sintomas:(82)(83)(84):
    - Instabilidade hemodinâmica
    - Suspeita de diagnósticos que necessitem de cuidados intensivos (ex.: dissecção da aorta)
    - Nova arritmia instável ou sintomática ou achado no ECG (por exemplo, taquicardia ventricular, fibrilação ventricular, bloqueio cardíaco avançado)
    - Síncope ou quase-síncope
    - Edema pulmonar provavelmente causado por isquemia.
    - Sopro de regurgitação mitral novo ou agravado, terceira bulha (B3) ou estertores.
  - Infarto agudo do miocárdio ou angina instável com complicações, conforme indicado por **um ou mais** dos seguintes critérios:(50)(81)(82)(83)(84)(85)(86):
    - Parada cardíaca
    - Dor torácica persistente
    - Instabilidade hemodinâmica
    - Nova arritmia instável ou sintomática ou achado no ECG (por exemplo, taquicardia ventricular, fibrilação ventricular, bloqueio cardíaco avançado)
    - Síncope ou quase-síncope
    - Edema pulmonar provavelmente causado por isquemia.
    - Sopro de regurgitação mitral novo ou agravado, terceira bulha (B3) ou estertores.
    - Bloco de ramificação de pacote de início recente
    - Novo bloqueio atrioventricular de alto grau
    - Insuficiência respiratória (por exemplo, nova necessidade de ventilação mecânica invasiva ou não invasiva)
    - Complicação hemorrágica (ex.: sangramento intracraniano ou no local de acesso após trombólise ou terapia antiplaquetária)
    - Alteração do estado mental grave ou persistente
    - Insuficiência renal aguda (estágio 3 da lesão renal aguda)
  - Complication of cardiac ablation, as indicated by **1 or more** of the following(87)(88)(89):
    - Pericardial tamponade
    - Hemodynamic instability
    - Thromboembolic stroke
    - Aortic or mitral valve injury
    - Vascular injuries
    - Esophageal perforation

    - Severe arrhythmia
    - Air embolism
    - Pneumothorax
    - Hemorrhage or hemothorax
    - Other severe complication
- Need for invasive hemodynamic monitoring with indwelling pulmonary artery catheter, as indicated by **1 or more** of the following[G](51)(52)(80)(91)(92):
    - Ablation with need for postprocedure invasive hemodynamic monitoring(87)
    - Suspected severe heart failure with unclear diagnosis (eg, pulmonary edema with left ventricular ejection fraction greater than 40%)(80)
    - Severe heart failure with ongoing symptoms or uncertain fluid status following initial diuretic treatment(80)
    - Cardiogenic shock due to heart failure(91)
    - Hemodynamic guidance needed for device selection and utilization of mechanical circulatory support(51)(52)
    - Severe pulmonary hypertension with need for titration of pulmonary vasodilator or inotropic therapy(93)
- Hemodynamic instability due to cardiac cause (eg, valve disease, arrhythmia, ischemia, conduction abnormality)(94)
- Hypertensive emergency (eg, systolic blood pressure greater than 180 mm Hg or diastolic blood pressure greater than 120 mm Hg with end organ damage), with need for **1 or more** of the following(95)(96):
    - IV antihypertensive therapy
    - Invasive hemodynamic monitoring (eg, arterial catheter)
- Infective endocarditis with **1 or more** of the following(97)(98)(99)(100):
    - Hemodynamic instability
    - Valvular dysfunction with heart failure
    - Need for inotropic agent
    - Severe arrhythmia
    - Intracranial mycotic aneurysm
    - Hemorrhagic stroke
    - Altered mental status that is severe or persistent
    - Acute respiratory failure
    - Requirement for frequent hemodynamic measurements
    - Acute kidney injury (stage 2)
    - Acute renal failure (stage 3 acute kidney injury)
    - Acute ischemic end organ damage (eg, small bowel infarction)
    - Need for acute surgical valve repair or replacement
    - Severe pulmonary edema (eg, severe mitral regurgitation)
- Pericardial tamponade(101)(102)
- Acute hemopericardium(101)
- Severe heart failure, as indicated by **1 or more** of the following(80)(103)(104)(105)(106)(107):
    - Hemodynamic instability
    - Respiratory failure
    - Severe arrhythmias
    - Evidence of cardiac ischemia
    - Acute kidney injury (stage 2)
    - Acute renal failure (stage 3 acute kidney injury)
    - Need for intravenous inotropic or vasopressor agent[H]
    - Need for mechanical circulatory support (eg, left ventricular assist device, intra-aortic balloon pump)(51)
- Myocarditis with **1 or more** of the following(109)(110)(111)(112):
    - Hemodynamic instability

        - Respiratory failure
        - Severe arrhythmias
        - Need for cardiac assist device (eg, left ventricular assist device or extracorporeal membrane oxygenator)
    - Syncope with **1 or more** of the following(113):
        - Hemodynamic instability
        - Severe arrhythmia or ECG finding (eg, ventricular tachycardia, ventricular fibrillation, advanced heart block)
        - Suspicion of diagnoses needing ICU care (eg, acute aortic dissection, massive pulmonary embolism, internal hemorrhage, critical aortic stenosis)
    - Electrical storm, as indicated by **1 or more** of the following(114):
        - 3 or more episodes of sustained ventricular tachycardia within 24 hours
        - Patient has implantable cardioverter-defibrillator (ICD) with 3 or more ICD interventions (shocks or antitachycardia pacing) within 24 hours.
    - Acute aortic syndrome(115)
    - Status post cardiac arrest(116)

- **Cardiovascular Surgery** diagnoses or procedures, including **1 or more** of the following(117):
    - Acute aortic dissection
    - Acute limb ischemia with need for catheter-directed thrombolysis(118)
    - Aortic surgery for **1 or more** of the following:
        - Thoracic aneurysm
        - Abdominal aneurysm with **1 or more** of the following(117)(119):
            - Emergency repair
            - Severe cardiopulmonary disease
            - Dialysis-dependent renal failure
            - Need for IV blood pressure control
            - Need for ongoing ventilatory support
            - Abdominal compartment syndrome
            - Perioperative complications, including **1 or more** of the following:
                - Hemodynamic instability
                - Cardiac ischemia or arrhythmia
                - Hypothermia
                - Blood transfusion greater than 3 L
                - Acute lower extremity ischemia
                - Ischemic bowel
                - Acute kidney injury (stage 2)
                - Acute renal failure (stage 3 acute kidney injury)
        - Aortic coarctation operative excision or repair
        - Aortofemoral or aortoiliac bypass with **1 or more** of the following(120):
            - Continued intubation and mechanical ventilation
            - Hemodynamic instability
            - Need for IV blood pressure control
            - Severe cardiopulmonary disease (eg, symptomatic heart failure, COPD, coronary artery disease)
            - Post-reperfusion compartment syndrome(121)
    - Cardiac surgery (eg, CABG, surgical valve replacement, open septal defect repair)
    - Periprocedural complication of transcatheter aortic valve replacement, as indicated by **1 or more** of the following(122)(123)(124):
        - Pericardial tamponade
        - Myocardial perforation
        - Severe arrhythmia

        - Hemodynamic instability
        - Periprocedural thromboembolic stroke
        - Hemorrhage or hemothorax
        - Acute kidney injury (stage 2)
        - Acute renal failure (stage 3 acute kidney injury)
    - Carotid endarterectomy or stent placement with **1 or more** of the following(117)(120)(125)(126)(127):
        - Blood pressure less than 100/60 mm Hg or greater than 160/90 mm Hg despite 4 hours of postanesthetic management
        - Need for IV blood pressure control
        - New or progressive neurologic defect
        - Suspected or proven postprocedure cerebral hyperperfusion syndrome, including **1 or more** of the following[I](127):
            - New-onset headache
            - Facial or eye pain
            - Altered mental status
            - Seizure
            - Intracerebral hemorrhage
            - Cerebral edema on brain imaging
        - Chest pain
        - Continued intubation
        - Heart failure
        - Airway compromise by hematoma or vocal cord paralysis
    - Heart transplant
    - Infrainguinal peripheral vascular surgery with **1 or more** of the following:
        - Hemodynamic instability
        - Acute complications such as persistent chest pain or respiratory distress
        - Requirement for IV antiarrhythmic or vasoactive agent
        - Requirement for pulmonary artery catheter
        - Severe hypertension despite 6 hours of recovery room management
        - Post-reperfusion compartment syndrome(121)
    - Percutaneous revascularization of the lower extremity with post-reperfusion compartment syndrome(121)
    - Complications of any cardiovascular surgery requiring ICU intervention, as indicated by **1 or more** of the following(128):
        - Hemodynamic instability
        - MI with complications (eg, severe arrhythmia, hypotension)
        - Excessive bleeding or severe coagulopathy
        - Respiratory failure
        - Acute kidney injury (stage 2) or Acute renal failure (stage 3 acute kidney injury)
        - Airway instability or obstruction
        - Neurologic deterioration
        - Infection with likelihood of sepsis syndrome or significant fluid shifts
- **Endocrinology** diagnoses or procedures, including **1 or more** of the following(129)(130):
    - Adrenal crisis (acute adrenal insufficiency) with **1 or more** of the following[J](131)(132)(133)(134)(135)(136)(137):
        - Hemodynamic instability
        - Altered mental status that is severe or persistent
        - Refractory hypoglycemia requiring glucagon or continuous glucose infusion with frequent adjustment
        - Precipitating factor requiring ICU care (eg, sepsis, trauma, hemorrhagic adrenalitis)
    - Pheochromocytoma or paraganglioma with **1 or more** of the following(136)(138)(139)(140):

    - Hypertensive emergency (eg, systolic blood pressure greater than 180 mm Hg or diastolic blood pressure greater than 120 mm Hg with end organ damage)
    - Need for IV vasoactive therapy
    - Need for invasive arterial pressure monitoring
    - Hypotensive shock
    - Acute myocardial ischemia
    - Cardiac arrhythmias of immediate concern (141)
    - Acute pulmonary edema
    - Other life-threatening end organ ischemia (eg, vasoconstriction induced stroke, mesenteric bowel ischemia, Acute kidney injury (stage 2), Acute renal failure (stage 3 acute kidney injury))
    - Postoperative for 24 hours after resection of pheochromocytoma
    - Postoperative Hemodynamic instability
  - Diabetic hyperosmolar state with **1 or more** of the following(142)(143)(144):
    - Altered mental status that is severe or persistent
    - Osmolality greater than 350 mOsm/kg (mmol/kg)[K]
    - Severe hypernatremia (eg, greater than 160 mEq/L (mmol/L))
    - Requirement for nursing care (eg, glucose checks) beyond capabilities of lower levels of care
  - Diabetic ketoacidosis with **1 or more** of the following(142)(145)(146)(147):
    - Serum pH less than 7.0 (adult) or 7.1 (child)
    - Serum bicarbonate less than 10 mEq/L (mmol/L)
    - Rapidly changing electrolytes
    - Significant hypokalemia (eg, ECG changes)
    - Hypotension
    - Requirement for large-volume fluid resuscitation
    - Requirement for nursing care (eg, glucose checks) beyond capabilities of lower levels of care
    - Respiratory insufficiency
    - Life-threatening cardiac dysrhythmias
    - Altered mental status that is severe or persistent
    - Acute kidney injury (stage 2)
    - Acute renal failure (stage 3 acute kidney injury)
    - Severe precipitating condition such as sepsis, stroke, or acute MI
    - Child at increased risk of cerebral edema (eg, age younger than 5 years, high BUN)
    - Child with clinical signs of cerebral edema (eg, headache, Altered mental status, cranial nerve palsy, papilledema, rising blood pressure with slowing heart rate)
    - Child with PCO_2 less than 21 mm Hg (2.8 kPa)
  - Severe hypoglycemia with **1 or more** of the following(145)(148):
    - Hemodynamic instability
    - Continuous glucose infusion with frequent adjustment required (eg, in response to glucose checks)
    - Glucagon administration required
  - Hyperthyroidism associated with thyroid storm (also known as thyrotoxic crisis), as indicated by **1 or more** of the following(135)(149)(150)(151)(152):
    - Score of 45 or more on the Burch-Wartofsky Point Scale[L]
    - Altered mental status that is severe or persistent
    - Heart rate greater than 130 beats per minute(150)
    - Cardiac arrhythmias of immediate concern
    - Temperature of 103.0 degrees F (39.5 degrees C) or more
    - Pulmonary edema
    - Hemodynamic instability [M](153)

- Decompensated hypothyroidism with **1 or more** of the following(135)(154):
  - Severe precipitating factor (eg, sepsis, myocardial infarction, stroke)
  - Hemodynamic instability
  - Altered mental status that is severe or persistent (eg, "myxedema coma")
  - Cardiac arrhythmias of immediate concern
  - Impending or actual pericardial tamponade
  - Status epilepticus attributable to hypothyroidism
  - Respiratory failure or airway compromise requiring, or likely to require mechanical ventilation
  - Rhabdomyolysis requiring large-volume IV fluid infusion
  - Acute renal failure (stage 3 acute kidney injury)
  - Other life-threatening end organ or electrolyte sequelae
- Arginine vasopressin deficiency (central diabetes insipidus (DI)) or arginine vasopressin resistance (nephrogenic DI) with polyuria that cannot be controlled rapidly (eg, with a vasopressin analogue for arginine vasopressin deficiency, or amiloride for arginine vasopressin resistance) requiring large-volume resuscitation and frequent urine output and serum sodium measurements(155)(156)
- Pituitary apoplexy[N](129)(157)(158)
- Thyroidectomy or parathyroidectomy with postoperative upper airway compromise due to postoperative bilateral vocal cord palsy, hematoma, or neck swelling(159)(160)
- Complications of any endocrine surgery requiring ICU intervention, as indicated by **1 or more** of the following(161):
  - Hemodynamic instability
  - MI with complications (eg, severe arrhythmia, Hypotension)
  - Excessive bleeding or severe coagulopathy
  - Respiratory failure or insufficiency
  - Renal failure
  - Airway instability or obstruction
  - Neurologic deterioration
  - Infection with Hypotension or significant fluid shifts

Gastroenterology diagnoses or procedures, including **1 or more** of the following(162):

- Acute diverticulitis with **1 or more** of the following(163)(164):
  - Active GI bleeding (eg, transfusion requirement greater than 2 units of packed red blood cells)
  - Coagulopathy
  - Hemodynamic instability
  - Vital sign abnormality due to infection (eg, peritonitis)
  - Sepsis with end organ dysfunction or failure(5)
- Esophageal or gastric perforation(165)
- Severe caustic esophageal or gastric injury (eg, endoscopic findings of necrosis or perforation, concomitant airway injury with risk of upper airway obstruction)(162)(166)(167)(168)
- Liver disease complications with **1 or more** of the following(169)(170)(171)(172)(173)(174)(175)(176)(177):
  - Severe hepatic encephalopathy (eg, stage 3 (somnolent) or higher)
  - Cerebral edema
  - Type 1 hepatorenal syndrome
  - Other cirrhosis-associated causes of renal failure (eg, severe hypovolemia, acute tubular necrosis, abdominal compartment syndrome)
  - Hemodynamic instability
  - Respiratory insufficiency due to severe ascites
  - Severe electrolyte abnormalities
  - Vital sign abnormality or end organ failure due to infection (eg, bacterial peritonitis)
  - Need for hemodynamic monitoring (eg, comorbid heart failure, valvular heart disease)
  - Need for mechanical ventilation

- Acute liver failure when aggressive intervention or transplant is anticipated(173)(178)(179)
- Gallbladder or bile duct inflammation (eg, cholangitis or cholecystitis) with **1 or more** of the following(180)(181)(182)(183):
  - Concomitant severe pancreatitis
  - Hemodynamic instability
  - Sepsis with end organ dysfunction or failure(5)(184)
- Gastroenteritis with **1 or more** of the following(5)(185):
  - Hemodynamic instability
  - Altered mental status that is severe or persistent
  - Toxic megacolon
  - Unstable comorbid condition
  - Need for hemodynamic monitoring (eg, comorbid heart failure, valvular heart disease)
  - Need for vasopressor agent
  - Acute end organ dysfunction or failure (eg, worsening renal function due to hemolytic uremic syndrome, sepsis, or severe dehydration)(5)(184)
- Gastrointestinal hemorrhage (upper or lower) with **1 or more** of the following(163)(186)(187)(188)(189)(190)(191)(192):
  - Active ongoing bleeding
  - Transfusion requirement greater than 2 units of packed red blood cells
  - Child with greater than 20% of blood volume loss
  - Bleeding ulcer, visible blood vessel, or bleeding (or recently bleeding) esophageal varices seen on endoscopy
  - Hypotension
  - Syncope
  - Coagulopathy
  - Hepatic cirrhosis
  - Altered mental status
  - Unstable comorbid condition or end organ dysfunction
  - Ischemia due to poor perfusion
  - Transfusion-related acute lung injury (TRALI) or transfusion-related circulatory overload (TACO)
  - Need for invasive (eg, pulmonary artery catheter) hemodynamic monitoring (eg, for patient with severe heart failure or valvular disease with uncertain fluid status)
  - Need for intravenous vasopressor agent
  - Need for intubation and mechanical ventilation (eg, for airway protection due to excessive hematemesis, or for hemodynamic instability)
- Gastrointestinal obstruction with **1 or more** of the following(193):
  - Hypotension
  - Need for aggressive fluid resuscitation
  - Evidence for bowel ischemia (eg, metabolic lactic acidosis, absent bowel sounds)
  - Bowel perforation
  - Sepsis with end organ dysfunction or failure(5)(184)
- Severe pancreatitis, as indicated by **1 or more** of the following(194)(195):
  - Requirement for aggressive fluid resuscitation
  - Severe electrolyte abnormality
  - Hypotension
  - Need for intravenous vasopressor agent
  - Persistent tachycardia greater than 120 beats per minute
  - Patient at high risk of rapid deterioration, including **1 or more** of the following:
    - Calculated Apache II score greater than 8[O]
    - Age older than 55 years
    - BMI greater than 30 BMI Calculator

        - Greater than 30% pancreatic necrosis on CT scan
        - Admission hematocrit greater than 47% (0.47)
    - Organ failure, as indicated by **1 or more** of the following:
        - Serum creatinine greater than 1.9 mg/dL (168 micromoles/L)
        - Urine output less than 50 mL/hour
        - Requirement for mechanical ventilation
        - Arterial partial pressure of oxygen less than 60 mm Hg (8.0 kPa) despite supplemental oxygen
        - PaO_2/FiO_2 ratio less than 300 mm Hg
    - Expanding pseudocyst
    - Infected pancreas
    - Need for pain control (eg, IV opioids) not performable at lower level of care
    - Pleural effusion
    - Encephalopathy
    - Severe active comorbidities (eg, liver disease)
    - Severe deterioration following admission to floor or intermediate unit, as evidenced by **1 or more** of the following(195)(196)(197):
        - Worsening renal failure (eg, rising BUN or creatinine, urine output less than 0.5 mL/kg/hour for more than 6 hours)
        - Persistent systemic inflammatory response syndrome (SIRS) criteria (temperature below 96.8 degrees F (36 degrees C) or above 100.4 degrees F (38 degrees C), pulse greater than 90 beats per minute, respiratory rate greater than 20 breaths per minute, white blood cell count lower than 4000/mm^3 (4 $x10^9$/L) or greater than 12,000/mm^3 (12 $x10^9$/L)) despite adequate fluid resuscitation
        - C-reactive protein greater than 15 mg/dL (150 mg/L)
        - Imaging findings of pancreatic or peripancreatic necrosis consistent with evolving severe pancreatitis
        - Worsening hypoxia, or pulmonary complications requiring invasive or noninvasive ventilation
- Toxic megacolon (eg, from inflammatory bowel disease, ischemic bowel, infection)(198)(199)
- Abdominal compartment syndrome requiring close monitoring or intervention, as indicated by **ALL** of the following(200):
    - Sustained intra-abdominal pressure (eg, bladder pressure) greater than 20 mm Hg
    - New organ dysfunction or failure (eg, respiratory failure, Hemodynamic instability, Acute kidney injury (stage 2), Acute renal failure (stage 3 acute kidney injury), intestinal ischemia) attributable to increased intra-abdominal pressure

General Surgery diagnoses or procedures, including **1 or more** of the following(7)(128)(161):

- Acute abdominal catastrophe (eg, ischemic bowel, perforated viscus, abdominal compartment syndrome)
- Sepsis with end organ failure (eg, from abdominal abscess, infected gallbladder, or diverticulum)(5)(11)
- Complications of any surgery requiring ICU intervention, as indicated by **1 or more** of the following:
    - Hemodynamic instability
    - MI with complications (eg, severe arrhythmia, Hypotension)
    - Excessive bleeding or severe coagulopathy
    - Respiratory failure or insufficiency
    - Acute renal failure (stage 3 acute kidney injury)
    - Airway instability or obstruction
    - Neurologic deterioration
- Abdominal trauma with **1 or more** of the following(201)(202)(203):
    - Hemodynamic instability
    - Uncontrolled bleeding
    - Need for intubation and mechanical ventilation
    - High-grade splenic injury (eg, ruptured hematoma, splenic vascular injury)
    - Moderate to severe hepatic injury (American Association for the Surgery of Trauma (AAST) injury scoring scale grade III or higher, major hepatic vein injuries or avulsion)
    - Following catheter-directed embolization of abdominal organ bleed

        - Following damage-control surgery
        - Abdominal compartment syndrome requiring close monitoring or intervention, as indicated by **ALL** of the following(203)(204):
            - Sustained intra-abdominal pressure (eg, bladder pressure) greater than 20 mm Hg
            - New organ dysfunction or failure (eg, respiratory failure, Hemodynamic instability, Acute kidney injury (stage 2), Acute renal failure (stage 3 acute kidney injury), intestinal ischemia) due to increased intra-abdominal pressure
        - Child with liver or spleen injury with Vital sign abnormality despite initial volume resuscitation(205)
    - Multiple trauma with complicating features, as indicated by **1 or more** of the following[P](201)(206)(207):
        - Serious injury involving more than one organ or system
        - Single organ injury requiring critical care intervention or monitoring (eg, invasive hemodynamic monitoring, frequent vital signs or neurologic checks)
        - Impending acute respiratory failure due to lung contusion, unstable chest wall, aspiration, hemorrhage, tension or open pneumothorax, or fat embolism
        - Facial or neck injury threatening airway patency
        - Cardiac contusion
        - Pericardial effusion
        - Bronchial tear
        - Hemodynamic instability
        - Rhabdomyolysis requiring large-volume IV fluid resuscitation
        - Compartment syndrome
        - Other significant complicating feature
    - Organ transplant(208)(209)(210)
    - Esophagectomy with **1 or more** of the following(7)(128)(211)(212):
        - Need for postoperative mechanical ventilation
        - Hemodynamic instability
        - Postoperative complication requiring ICU intervention (eg, myocardial infarction, excessive bleeding, Acute renal failure (stage 3 acute kidney injury))
    - Pancreatectomy (eg, pancreaticoduodenectomy, also known as Whipple procedure)[Q](196)
    - Necrotizing soft tissue infection (eg, necrotizing fasciitis, *Clostridial* gas gangrene, Fournier gangrene)(213)(214)(215)
    - Preoperative or postoperative patient requiring ICU intervention, such as hemodynamic optimization, pulmonary artery monitoring, mechanical ventilation, or extensive nursing care
    - Bariatric surgery patient with **1 or more** of the following(216)(217):
        - ICU management needs for comorbid conditions (eg, airway issues due to severe sleep apnea)
        - Failed postoperative extubation
        - Intraoperative complications (eg, perforated viscus, bleeding)
- **Head and Neck Surgery** diagnoses or procedures, including **1 or more** of the following(218)(219)(220):
    - Complications of any surgery requiring ICU intervention, as indicated by **1 or more** of the following(161):
        - Hemodynamic instability
        - MI with complications (eg, severe arrhythmia, Hypotension)
        - Excessive bleeding or severe coagulopathy
        - Respiratory failure or insufficiency
        - Renal failure
        - Airway instability or obstruction
        - Neurologic deterioration
        - Infection with likelihood of sepsis syndrome or significant fluid shifts
    - Preoperative or postoperative patient requiring ICU intervention, such as pulmonary artery monitoring, mechanical ventilation, or extensive nursing care
    - Life-threatening infection of head and neck, including **1 or more** of the following(215)(221)(222)(223)(224)(225):
        - Ludwig angina (submandibular and lingual cellulitis)
        - Parapharyngeal (lateral) or retropharyngeal (posterior) infection or abscess
        - Peritonsillar abscess

        - Lemierre syndrome (*Fusobacterium* oropharyngeal infection complicated by jugular vein septic thrombophlebitis)
        - Acute epiglottitis
        - Rhino-orbital-cerebral mucormycosis(226)
        - Other upper airway infection causing or threatening airway compromise
    - Airway or hemodynamic compromise that persists after 3 hours of observation in postanesthesia care unit following nasal, palate (eg, uvulopalatopharyngoplasty or palatoplasty), or tongue surgery for sleep apnea
    - Symptomatic upper airway compromise (eg, laryngeal edema, mass)(225)(227)
    - Massive epistaxis with hemodynamic or airway compromise(228)
    - Posterior nasopharyngeal bleed with upper airway compromise(228)
    - Other airway-compromising procedure

- **Hematology - Oncology** diagnoses or procedures, including chemotherapy administration, with **1 or more** of the following(229)(230)(231)(232)(233)(234)(235)(236)(237)(238)(239):
    - Hemodynamic instability
    - Neutropenic fever with sepsis and end organ failure or dysfunction(5)(11)(238)
    - Hemophilia with severe bleeding, as indicated by **1 or more** of the following(240)(241):
        - Hemodynamic instability
        - CNS bleed
        - Gastrointestinal bleeding with coagulopathy that is not quickly reversible (eg, with clotting factor replacement)
        - Pulmonary hemorrhage
        - Intramuscular bleed with compartment syndrome
        - Hematoma obstructing, or at risk of obstructing, airway (eg, retropharyngeal space or tongue)
        - Epistaxis with posterior nasal packing compromising upper airway
        - Need for intubation and mechanical ventilation (eg, for airway protection, transfusion-related lung injury, or circulatory overload)
    - Severe sickle cell crisis, as indicated by **1 or more** of the following(242)(243)(244):
        - Hemodynamic instability
        - Evidence of CNS injury (eg, stroke, encephalopathy)
        - Altered mental status that is severe or persistent
        - Respiratory distress
        - Severe hypoxemia (eg, SpO_2 less than 95% (or 3% below baseline) on more than 50% inspired oxygen)[C][R]
        - Need for invasive or noninvasive mechanical ventilation
        - End organ ischemia (eg, intestinal) or failure
        - Acute kidney injury (stage 2)
        - Acute renal failure (stage 3 acute kidney injury)
        - Hyperhemolysis
        - Need for treatment (eg, pain control with IV opioids) or monitoring (eg, aplastic crisis) not performable at lower level of care
    - Anemia with serious clinical findings, including **1 or more** of the following(245)(246)(247):
        - Hemodynamic instability
        - Uncontrolled active bleeding
        - Altered mental status that is severe or persistent
        - End organ ischemia or dysfunction (eg, myocardial or intestinal ischemia, Acute kidney injury (stage 2), Acute renal failure (stage 3 acute kidney injury))(247)
        - Cardiac arrhythmias of immediate concern
        - Need for invasive (eg, pulmonary artery catheter) hemodynamic monitoring (eg, for patient with concomitant severe heart failure or valvular disease with uncertain fluid status)
        - Unstable comorbid condition
    - Severe or life-threatening bleeding (eg, intracranial, major gastrointestinal, extensive mucosal, massive hemoptysis, diffuse alveolar hemorrhage)
    - Disseminated intravascular coagulation[S](248)(249)(250)(251)

- Platelet count less than 20,000/mm^3 (20 $x10^9$/L) with active bleeding
- Hypercalcemia of malignancy with **1 or more** of the following(252):
  - Calcium greater than 14 mg/dL (3.5 mmol/L)(31)
  - Calcium greater than 12 mg/dL (3 mmol/L) with **1 or more** of the following(31):
    - Altered mental status
    - Cardiac arrhythmia or conduction disturbance
    - Dehydration that is severe or persistent
- Syndrome of inappropriate antidiuresis (SIAD) due to malignancy or chemotherapy and **1 or more** of the following(24)(253):
  - Serum sodium less than 110 mEq/L (mmol/L)
  - Serum sodium less than 125 mEq/L (mmol/L) and **1 or more** of the following:
    - Altered mental status
    - Seizure
    - Confirmed or suspected elevated intracranial pressure (eg, on brain imaging or presence of Cushing reflex (bradycardia, wide pulse pressure, and irregular respirations))
- Leukostasis and **1 or more** of the following[T](229)(234)(239):
  - Respiratory manifestations (eg, dyspnea, Hypoxemia, acute respiratory distress syndrome)
  - Neurologic manifestations (eg, focal deficit, Altered mental status, seizure)
  - Cardiac ischemia
  - Other end organ dysfunction (eg, Acute kidney injury (stage 2), Acute renal failure (stage 3 acute kidney injury), visual disturbances)
- Malignancy-associated hemophagocytic lymphohistiocytosis[U](234)(254)
- Superior vena cava or superior mediastinal syndrome and **1 or more** of the following:
  - Hemodynamic instability
  - Respiratory symptoms (eg, dyspnea, stridor, Tachypnea)
  - Known or suspected pericardial effusion
  - Known or suspected airway compromise
  - Cerebral edema
  - Altered mental status
- Pericardial tamponade
- Hyperviscosity syndrome with **1 or more** of the following(234):
  - Altered mental status
  - Focal neurologic findings
  - Seizures
  - Mucosal bleeding (eg, epistaxis, gingival, gastrointestinal)
  - Visual abnormalities
  - Dyspnea
  - Heart failure
  - Myocardial ischemia
  - Limb ischemia
- Severe over-anticoagulation with active bleeding and **1 or more** of the following(255)(256):
  - Hemodynamic instability
  - Bleeding that compromises organ function (eg, pericardial tamponade, intracranial or spinal cord hemorrhage, intramuscular bleed with compartment syndrome)
- Thrombotic microangiopathy syndrome (eg, thrombotic thrombocytopenic purpura, hemolytic uremic syndrome, HELLP (hemolysis, elevated liver tests, low platelet level) syndrome of pregnancy) and **1 or more** of the following(251)(257)(258)(259)(260)(261):
  - Acute kidney injury (stage 2)
  - Acute renal failure (stage 3 acute kidney injury)

        - Altered mental status
        - Seizure
        - New-onset focal neurologic deficit
        - Active blood loss
        - Cardiac arrhythmia
        - Life-threatening bleed (eg, CNS bleed, pulmonary hemorrhage)
        - Other end organ dysfunction (eg, acute respiratory distress syndrome, ischemic bowel, cardiac ischemia, pancreatitis)
        - Precipitating severe infection requiring ICU care (eg, sepsis due to *Streptococcus pneumoniae*; Shiga-toxin-producing bacteria, including *Shigella dysenteriae*, *Escherichia coli*)
    - Immune thrombocytopenia (ITP) and **1 or more** of the following(262)(263):
        - Bleeding into a critical anatomic site (eg, intracranial, intraspinal, intraocular, pericardial bleed)
        - Bleeding with Hemodynamic instability
        - Uncontrolled active bleeding
        - Respiratory compromise
    - Thrombotic storm(264)(265)
    - Transfusion-related lung injury (TRALI), transfusion-associated circulatory overload (TACO), or transfusion-related anaphylaxis(266)(267)
    - Tumor lysis syndrome with **1 or more** of the following(268)(269)(270):
        - Acute kidney injury (stage 2)
        - Acute renal failure (stage 3 acute kidney injury)
        - Severe electrolyte abnormality
        - Cardiac dysrhythmia
        - Seizure
    - Tumor-associated status epilepticus(271)(272)
    - End organ failure due to immune checkpoint blockade-related adverse events
    - Cytokine release syndrome (eg, following blinatumomab or chimeric antigen receptor (CAR) T-cell therapy) with **1 or more** of the following(238)(273)(274):
        - Hemodynamic instability
        - Signs of neurologic toxicity (eg, seizures, encephalopathy)
        - Need for intravenous vasopressor agent
        - Severe hypoxemia (eg, more than 6 L/min nasal oxygen flow needed to maintain SaO_2 greater than 90% or greater)[C]
        - Need for invasive or noninvasive mechanical ventilation
        - Acute kidney injury (stage 2)
        - Acute renal failure (stage 3 acute kidney injury)
    - Immune effector cell-associated neurotoxicity syndrome (eg, following chimeric antigen receptor (CAR) T-cell therapy) with **1 or more** of the following(273):
        - Hemodynamic instability
        - Hypoxia requiring noninvasive or invasive mechanical ventilation
        - Altered mental status that is severe or persistent
        - Focal neurologic finding
        - Seizure
        - Cerebral edema on brain imaging study
    - Lambert-Eaton or other paraneoplastic syndrome with respirator muscle weakness requiring frequent monitoring of pulmonary vital capacity, noninvasive ventilation, or mechanical ventilation(271)
- **Infectious Disease** diagnosis, with **1 or more** of the following(1)(5)(11)(215)(275)(276):
    - Hemodynamic instability
    - Requirement for frequent hemodynamic measurements (eg, arterial catheter, pulmonary artery catheter)
    - End organ dysfunction (eg, Acute kidney injury (stage 2), Acute renal failure (stage 3 acute kidney injury), respiratory failure)

- Necrotizing soft tissue infection (eg, necrotizing fasciitis, clostridial myonecrosis, clostridial gas gangrene, Fournier gangrene)(214)(215)(277)(278)
- Deep neck infection (eg, Ludwig angina, parapharyngeal (lateral) or retropharyngeal (posterior) space infection, peritonsillar abscess)[V](215)(225)
- Acute epiglottitis(222)(225)(279)
- Rhino-orbital cerebral mucormycosis(226)
- Toxic shock syndrome (*Staphylococcus aureus* or *Streptococcus pyogenes* infection with fever, diffuse macular rash, hypotension, and multiorgan failure)[W](145)
- Tetanus (for ventilator support, control of rigidity with benzodiazepines or neuromuscular blocking agent, treatment of severe autonomic dysfunction including alternating hypertension, Hemodynamic instability, Tachycardia, and Bradycardia)(280)
- Botulism (for respiratory monitoring and ventilator support and to monitor for anaphylaxis following antitoxin administration)(281)
- Rabies(282)
- Severe malaria, as indicated by **1 or more** of the following(283)(284)(285):
  - Hemodynamic instability
  - Altered mental status that is severe or persistent
  - Glasgow coma score of 10 or less (or Blantyre coma score of 3 or less for preverbal child) GCS Calculator
  - Neuroimaging showing cerebral edema or hemorrhage
  - Seizure with more than 2 episodes over 24 hours, status epilepticus, or leading to airway compromise requiring (or likely to require) mechanical ventilation[X](286)(287)
  - Metabolic acidosis (eg, serum bicarbonate less than 15 mEq/L (mmol/L) or lactate greater than 45 mg/dL (5 mmol/L))
  - Severe hypoglycemia requiring continuous glucose infusion with frequent adjustment or glucagon administration(145)
  - Acute respiratory distress syndrome (noncardiogenic pulmonary edema)
  - Severe respiratory findings
  - Disseminated intravascular coagulopathy (DIC)
  - Other major organ impairment (eg, Acute renal failure (stage 3 acute kidney injury), liver failure)
  - Significant bleeding (eg, hematemesis, hemoptysis or alveolar hemorrhage, melena)
- Viral hemorrhagic fever(288)
- Venom exposure from bite or sting and **1 or more** of the following(289)(290):
  - Moderate to severe envenomation requiring antivenin therapy[Y]
  - Hemodynamic instability
  - Neurotoxicity causing weakness, Altered mental status, impaired respiration, or seizures (eg, from Elapidae (coral, cobra, mamba) snake bite, Mojave rattlesnake)
  - Coagulopathy with uncontrolled bleeding
  - Pulmonary edema
  - Severe respiratory findings
  - Severe airflow or ventilation abnormalities
  - Acute kidney injury (stage 2), or Acute renal failure (stage 3 acute kidney injury)
- Orbital cellulitis with **1 or more** of the following(291)(292)(293):
  - New neurologic signs (eg, cranial nerve dysfunction, Altered mental status)
  - Meningeal signs
  - Acute visual impairment or change in color vision
  - Hemodynamic instability
  - Need for invasive hemodynamic monitoring (eg, arterial or pulmonary artery catheter)
  - Immunosuppressed patient
  - Cavernous sinus thrombosis

Nephrology diagnoses or procedures, including acute or acute on chronic renal insufficiency, with **1 or more** of the following(294)(295)(296)(297):

- Severe electrolyte or acid-base disorder
- Acute pulmonary edema
- Hypotension

- Hypertensive emergency (eg, systolic blood pressure greater than 180 mm Hg or diastolic blood pressure greater than 120 mm Hg with end organ damage), with need for **1 or more** of the following(95)(96):
  - IV antihypertensive therapy (eg, nicardipine, labetalol)
  - Invasive hemodynamic monitoring (eg, arterial catheter)
- Underlying critical illness contributing to renal failure (eg, septic shock, type 1 hepatorenal syndrome)(298)(299)(300)
- Need for continuous renal replacement therapy[Z](57)(294)
- Pyelonephritis with Hemodynamic instability (5)(11)(275)

Neurology diagnoses or procedures, including **1 or more** of the following(301)(302)(303)(304):

- Intracranial hypertension requiring **1 or more** of the following(305)(306)(307):
  - Induced barbiturate coma
  - Pharmacologic paralysis or deep sedation and mechanical ventilation
  - Intracranial pressure or cerebral perfusion pressure monitoring
  - IV hyperosmolar therapy (eg, hypertonic saline (preferred) or mannitol)
  - Frequent serum osmolality measurements
  - IV vasoactive medication (eg, labetalol for hypertension, norepinephrine for hypotension)
  - Ventriculostomy drainage
  - Therapeutic hypothermia
  - Surgical decompression
- Seizures with **1 or more** of the following(287)(308)(309)(310):
  - Status epilepticus
  - Severe electrolyte abnormalities causing seizures
  - Airway compromise requiring or likely to require mechanical ventilation
- Progressive acute neurologic dysfunction requiring or likely to require **1 or more** of the following(311)(312)(313)(314)(315):
  - Mechanical ventilation
  - Intracranial pressure or cerebral perfusion pressure monitoring
- Guillain-Barre syndrome (variants include acute inflammatory demyelinating polyradiculoneuropathy (AIDP), acute motor axonal neuropathy (AMAN), or Miller Fisher syndrome) with **1 or more** of the following(313)(316)(317):
  - Progressive weakness (risk for respiratory deterioration and need for intubation and mechanical ventilation)
  - Progressive decrease in respiratory function (eg, vital capacity, maximum inspiratory or expiratory pressure)
  - Fall in vital capacity of more than 30% below the predicted baseline over 24 hours
  - Single breath count less than 20 (inability to count from 1 to 20 in a single breath)
  - Bulbar muscle involvement (risk for aspiration and need for intubation and mechanical ventilation)
  - Modified Erasmus GBS respiratory insufficiency scale score of 25 or higher[AA]
  - Severe autonomic dysregulation, including **1 or more** of the following:
    - Hypertensive crisis
    - Severe hypertension
    - Bradycardia
    - Tachycardia
    - Cardiac arrhythmias of immediate concern
- Myasthenia gravis crisis with **1 or more** of the following(313)(318)(319):
  - Respiratory insufficiency requiring, or likely to require, intubation and mechanical ventilation[BB]
  - Bulbar muscle involvement (risk for aspiration and need for intubation and mechanical ventilation)
- CNS infection (eg, meningitis, encephalitis, intracranial abscess) with **1 or more** of the following[CC](215)(322)(323)(324)(325)(326)(327):
  - Hemodynamic instability
  - Altered mental status that is severe or persistent

    - Respiratory insufficiency
    - Rapidly evolving rash
    - Elevated intracranial pressure
    - Hydrocephalus
    - Refractory seizures
    - Focal neurologic deficits
    - Agitation requiring sedation
    - Sepsis and end organ failure or dysfunction(5)(11)
- Stroke with **1 or more** of the following[DD](60)(328)(330)(331):
    - Need for observation after thrombolysis or endovascular treatment[EE]
    - Altered mental status
    - Need for mechanical ventilation
    - Elevated intracranial pressure
    - Hypertensive emergency
    - High risk of progressive infarction or deterioration based on CT scan or MRI
    - Large hemispheric or cerebellar infarct
    - Arterial dissection
    - New seizure
    - Hemorrhage
    - Child with acute stroke(332)
- Acute coma(333)
- Acute paraplegia or tetraplegia[FF](334)
- Acute spinal cord compression (eg, epidural abscess, hematoma, tumor)(272)(335)
- Acute spinal cord infarction(335)
- Acute flaccid paralysis[GG](335)(336)
- Acute spontaneous intracranial hemorrhage[HH](337)(338)(339)(340)(341)(342)
- Acute subarachnoid hemorrhage[II](343)
- Traumatic brain injury with **1 or more** of the following(344)(345)(346)(347)(348)(349):
    - Altered mental status that is severe or persistent or Glasgow coma scale score less than 10 GCS Calculator
    - Cerebral edema
    - Cerebral hemorrhage
    - Increased intracranial pressure
    - Need for ICP and cerebral perfusion pressure monitoring, tissue oxygen monitor, or ventriculostomy
    - Need for arterial catheter for blood pressure monitoring
    - Need for intubation and mechanical ventilation(350)
- Acute vestibular syndrome with **1 or more** of the following(328):
    - Posterior circulation stroke requiring thrombolytic therapy or acute endovascular surgery
    - Elevated intracranial pressure or edema at high risk of brainstem compression
    - Hemorrhagic conversion
- Drug ingestion or toxic exposure with **1 or more** of the following(351)(352)(353)(354):
    - Hemodynamic instability
    - Respiratory depression (eg, new partial pressure of carbon dioxide greater than 45 mm Hg (6.0 kPa))
    - Patient requires or is likely to require mechanical ventilation.
    - Arrhythmia
    - Seizures

        - Altered mental status that is severe or persistent
        - Significant risk for acute deterioration (eg, toxic drug level or ingestion of overdose of sustained-release preparation)
        - Drug-induced Hypothermia or hyperthermia(63)
        - Significant or worsening metabolic acidosis
        - Severe hypoglycemia requiring glucose infusion with frequent adjustment or glucagon administration beyond the capabilities of a lower level of care
        - Ongoing antidote administration (eg, continuous naloxone infusion, organophosphate toxicity treatment)
        - Acute kidney injury (stage 2) or Acute renal failure (stage 3 acute kidney injury)
        - Acute liver failure
        - Drug-induced rhabdomyolysis requiring high-volume IV fluid resuscitation
        - Need for pharmacologic restraints
        - Emergency intervention need (eg, dialysis, hemoperfusion, plasma exchange, restraints)
    - Parkinson hyperpyrexia syndrome, as evidenced by **1 or more** of the following after reduction or discontinuation of anti-Parkinsonian medication, or following direct brain stimulation discontinuation or malfunction(74):
        - Muscle rigidity
        - Hyperthermia
        - Altered mental status
        - Hemodynamic instability
        - Elevated creatine kinase
    - Status dystonicus (generalized sustained severe dystonic spasms with end organ decompensation or failure)[JJ](74)(355)
    - Cerebral venous thrombosis[KK](356)(357)(358)
    - Autoimmune or paraneoplastic encephalitis with **1 or more** of the following[LL](327)(359)(360):
        - Altered mental status that is severe or persistent
        - Hemorrhage
        - Elevated intracranial pressure
        - Refractory seizures
        - Respiratory insufficiency
        - Severe hyponatremia (eg, serum sodium less than 110 mEq/L (mmol/L) or serum sodium less than 125 mEq/L (mmol/L) with Altered mental status or seizure)
        - Severe autonomic dysregulation, including **1 or more** of the following:
            - Hemodynamic instability
            - Severe hypertension
            - Bradycardia
            - Tachycardia
            - Cardiac arrhythmias of immediate concern
    - Brain death with preparation for organ donation

- **Neurosurgery** diagnoses or procedures, including **1 or more** of the following(306)(361):
    - Emergency craniotomy for tumor, hematoma, hemorrhage, or trauma(272)(337)(344)(362)(363)(364)
    - Craniotomy for cerebral swelling associated with cerebral infarction unresponsive to medical therapy(306)(328)(365)
    - Elective craniotomy for posterior fossa tumor(366)
    - Elective craniotomy (supratentorial) for tumor with **1 or more** of the following(362)(367):
        - Rapid preoperative progression of focal deficit
        - Postoperative neurologic deficit or impaired consciousness 6 hours after completion of procedure
        - Systolic blood pressure less than 110 mm Hg or greater than 180 mm Hg despite therapy
        - Extensive operative blood loss
        - High anesthesia risk (eg, American Society of Anesthesiologists class greater than III). See American Society of Anesthesiologists (ASA) Physical Status Classification System.

        - Postoperative diabetes insipidus
        - Need for postoperative mechanical ventilation
    - Craniotomy for aneurysm with **1 or more** of the following(368):
        - Postoperative neurologic deficit or impaired consciousness 6 hours after completion of procedure
        - Preoperative Hunt-Hess grade 3 or higher
        - Rapid preoperative progression of focal deficit
        - Systolic blood pressure less than 110 mm Hg or greater than 180 mm Hg despite therapy
        - Intracranial pressure monitoring
        - Postoperative diabetes insipidus
        - Need for postoperative mechanical ventilation
    - Acute spinal cord injury(334)(369)
    - Subarachnoid hemorrhage(337)
    - Traumatic brain injury with **1 or more** of the following(344)(345)(346)(347)(348)(349)(370):
        - Altered mental status that is severe or persistent
        - Glasgow coma scale score less than 10 GCS Calculator
        - CT scan showing cerebral edema or hemorrhage
        - Need for intracranial pressure and cerebral perfusion pressure monitoring, tissue oxygen monitor, or ventriculostomy
        - Need for arterial catheter for blood pressure monitoring
        - Need for intubation or mechanical ventilation
    - Hypophysectomy with postoperative diabetes insipidus (arginine vasopressin deficiency) requiring large-volume resuscitation and frequent urine output and serum sodium measurements(156)(371)
    - Complications of surgery requiring ICU intervention, as indicated by **1 or more** of the following(161):
        - Hemodynamic instability
        - MI with complications (eg, severe arrhythmia, Hypotension)
        - Excessive bleeding or severe coagulopathy
        - Respiratory failure or insufficiency
        - Renal failure
        - Airway instability or obstruction
        - Neurologic deterioration
        - Infection with Vital sign abnormality
    - Preoperative or postoperative patient requiring ICU or neuro ICU intervention, such as pulmonary artery monitoring, mechanical ventilation, intracranial and cerebral perfusion pressure monitoring, tissue oxygen monitor, ventriculostomy, or intensive nursing care(161)
- **Obstetrics and Gynecology** diagnoses or procedures, including **1 or more** of the following(372)(373)(374)(375)(376)(377):
    - Severe peripartum condition, as indicated by **1 or more** of the following:
        - Eclampsia(376)
        - Hypertensive emergency (eg, systolic blood pressure greater than 160 mm Hg or diastolic blood pressure greater than 110 mm Hg with end organ damage), with need for IV antihypertensive therapy or invasive hemodynamic monitoring(376)(378)
        - HELLP (hemolysis, elevated liver enzymes, and low platelet count) syndrome(376)(379)(380)
        - Disseminated intravascular coagulation (DIC)[MM](250)
        - Pulmonary edema (eg, due to peripartum cardiomyopathy)(373)
        - Severe arrhythmia
        - Respiratory failure (eg, due to acute respiratory distress syndrome (ARDS) from eclampsia, sepsis, or amniotic fluid embolism, fluid overload, or respiratory depressant drugs (opioids, anesthetic agents) with underlying cardiopulmonary disease)(381)
        - Pulmonary embolism
        - Anaphylactoid syndrome of pregnancy (amniotic fluid embolus)(382)

- Severe or critical ovarian hyperstimulation syndrome, as indicated by **1 or more** of the following[NN](383):
  - Hypotension or Hemodynamic instability
  - Acute kidney injury (stage 2) or Acute renal failure (stage 3 acute kidney injury)
  - Respiratory distress or Severe respiratory findings (eg, due to massive hydrothorax or acute respiratory distress syndrome (ARDS))
  - Need for noninvasive or invasive mechanical ventilation
  - Cardiac arrhythmias of immediate concern
  - Pericardial tamponade
- Acute fatty liver of pregnancy (acute liver failure)(380)
- Vital sign abnormality due to infection (eg, puerperal sepsis, chorioamnionitis, septic abortion, pneumonia)
- Complications such as placental abruption or severe hemorrhage(384)
- Severe cardiac complication such as coronary or aortic dissection, or acute coronary syndrome
- Postpartum hypopituitarism (pituitary infarction due to peripartum hypotension, also known as "Sheehan syndrome") with refractory hypotension(157)
- Puerperal sepsis with end organ dysfunction (eg, Sequential Organ Function Assessment (SOFA) score increase of 2 or more from baseline) or Hemodynamic instability [OO](5)(184)(386)
- Diabetic ketoacidosis and **1 or more** of the following(143)(144)(146)(147):
  - Serum pH less than 7.0 (adult) or 7.1 (child)
  - Serum bicarbonate less than 10 mEq/L (mmol/L)
  - Rapidly changing electrolytes
  - Significant hypokalemia (eg, ECG changes)
  - Hypotension
  - Requirement for large-volume fluid resuscitation
  - Requirement for nursing care (eg, glucose checks) beyond capabilities of lower levels of care
  - Respiratory insufficiency
  - Life-threatening cardiac dysrhythmias
  - Altered mental status that is severe or persistent
  - Acute kidney injury (stage 2)
  - Acute renal failure (stage 3 acute kidney injury)
  - Severe precipitating condition such as sepsis, stroke, or acute MI

- Pelvic infection with sepsis and end organ dysfunction(5)(184)(275)
- Ruptured ectopic pregnancy
- Cerebral venous sinus thrombosis[PP](356)(357)(387)
- Complications of any surgery requiring ICU intervention, as indicated by **1 or more** of the following(161):
  - Hemodynamic instability
  - MI with complications (eg, severe arrhythmia, Hypotension)
  - Excessive bleeding or severe coagulopathy
  - Respiratory failure or insufficiency
  - Renal failure
  - Airway instability or obstruction
  - Neurologic deterioration
  - Infection with Vital sign abnormality
- Preoperative or postoperative patient requiring ICU intervention, such as pulmonary artery monitoring, mechanical ventilation, or extensive nursing care[QQ](161)(388)

Ophthalmology diagnoses or procedures, including **1 or more** of the following(389):

- Acute central retinal artery occlusion with need for observation following intravenous or catheter-directed thrombolytic therapy[RR](390)(391)
- Complications of any surgery requiring ICU intervention, such as **1 or more** of the following(161):
  - Hemodynamic instability

        - MI with complications (eg, severe arrhythmia, Hypotension)
        - Excessive bleeding or severe coagulopathy
        - Respiratory failure or insufficiency
        - Renal failure
        - Airway instability or obstruction
        - Neurologic deterioration
        - Infection with Vital sign abnormality
    - Preoperative or postoperative patient requiring ICU intervention, such as pulmonary artery monitoring, mechanical ventilation, or extensive nursing care(161)

- **Orthopedics** diagnoses or procedures, including **1 or more** of the following(392)(393):
    - Complications of any surgery requiring ICU intervention, as indicated by **1 or more** of the following(161):
        - Hemodynamic instability
        - MI with complications (eg, severe arrhythmia, Hypotension)
        - Excessive bleeding or severe coagulopathy
        - Respiratory failure or insufficiency
        - Renal failure
        - Airway instability or obstruction
        - Neurologic deterioration
        - Infection with Vital sign abnormality
    - Multiple trauma with complicating features, as indicated by **1 or more** of the following(201)(206)(207)(394):
        - Serious injury involving more than one organ or system
        - Single organ injury requiring critical care intervention or monitoring (eg, invasive hemodynamic monitoring, intracranial pressure monitoring, frequent vital signs or neurologic checks)
        - Impending acute respiratory failure due to lung contusion, unstable chest wall, aspiration, hemorrhage, tension or open pneumothorax, or fat embolism(395)
        - Facial or neck injury threatening airway patency
        - Cardiac contusion(396)(397)
        - Rhabdomyolysis requiring large-volume IV fluid resuscitation
        - Pericardial effusion
        - Bronchial tear
        - Hemodynamic instability
        - Cervical spine injury associated with respiratory insufficiency or neurogenic shock
        - Other significant complicating feature
    - Compartment syndrome diagnosed, suspected, or threatened(204)(398)(399)(400)
    - Surgical wound infection with **1 or more** of the following(5)(184)(275):
        - Hemodynamic instability
        - Necrotizing infection
        - Sepsis with end organ failure or dysfunction
        - Altered mental status that is severe or persistent
        - Requirement for invasive hemodynamic measurements (eg, from arterial line, central line, or pulmonary artery catheter)
    - Preoperative or postoperative patient requiring ICU intervention, such as pulmonary artery monitoring, mechanical ventilation, or extensive nursing care(161)

- **Thoracic Surgery and Pulmonary Disease** diagnoses or procedures, including **1 or more** of the following:
    - Asthma with **1 or more** of the following(401)(402)(403)(404)(405):
        - Respiratory distress
        - Impending or actual respiratory failure
        - Need for mechanical ventilation
        - Peak expiratory flow rate less than 30% of predicted or personal best

- Peak expiratory flow rate or forced expired volume in 1 second (FEV_1) less than 40% predicted after 1 hour of initial treatment
- Acidosis
- Persistent or worsening Hypoxemia after initial treatment
- Hypercapnia (eg, arterial partial pressure of carbon dioxide greater than 43 mm Hg (5.7 kPa))
- Altered mental status that is severe or persistent
- Need for continuous inhaled bronchodilator
- Cardiac arrhythmias of immediate concern
- Hypotension
- Barotrauma (pneumothorax, pneumoperitoneum, pneumopericardium)
- Child 5 years or younger and **1 or more** of the following:
  - Unable to speak or drink
  - Central cyanosis
  - Subcostal or subglottic contractions
  - Oxygen saturation less than 92%
  - Silent chest on auscultation
  - Respiratory rate greater than 40 breaths per minute
  - Age 0 to 3 years with pulse rate greater than 180 beats per minute
  - Age 4 to 5 years with pulse rate greater than 150 beats per minute

- Bronchiolitis in a child with **1 or more** of the following(406)(407):
  - Impending respiratory failure[SS]
  - Need for invasive or noninvasive ventilation
  - Recurrent apnea
  - Hypoxemia unresponsive to oxygen supplementation
- Bronchopulmonary dysplasia rehospitalization with **1 or more** of the following(408)(409):
  - Hemodynamic instability
  - Impending respiratory failure[SS]
  - Hypoxemia unresponsive to oxygen therapy
  - Need for invasive or noninvasive mechanical ventilation
  - Pulmonary hypertension with acute pulmonary hypertensive crisis
  - Need for invasive hemodynamic monitoring (eg, pulmonary artery or arterial catheter pressure monitoring)
  - Need for close observation and aggressive therapy (eg, suctioning, chest physiotherapy, or inhalation treatments) at intervals less than 1 hour
- COPD with **1 or more** of the following(410)(411):
  - Respiratory arrest
  - Need for mechanical ventilation[TT]
  - Hemodynamic instability
  - Cardiac arrhythmias of immediate concern
  - Severe dyspnea unresponsive to initial treatment
  - Altered mental status
  - Massive aspiration or persistent vomiting
  - Persistent findings despite oxygen and outpatient management, including **1 or more** of the following:
    - Arterial partial pressure of oxygen less than 40 mm Hg (5.3 kPa)
    - Arterial partial pressure of carbon dioxide greater than 60 mm Hg (8.0 kPa) and uncompensated respiratory acidosis (pH less than 7.35)
    - Severe uncompensated acidosis (pH less than 7.25)
    - Worsening Hypoxemia or acidosis
- Cor pulmonale with **1 or more** of the following(93)(410)(412)(413)(414):

    - Respiratory arrest
    - Hemodynamic instability
    - Need for IV inotropic or vasoactive agent
    - Need for inhaled nitric oxide or epoprostenol
    - Need for invasive hemodynamic monitoring (eg, central venous, pulmonary artery, or arterial catheter)
    - Arterial partial pressure of oxygen less than 40 mm Hg (5.3 kPa)
    - Worsening Hypoxemia or acidosis despite oxygen therapy
    - Unstable atrial tachyarrhythmia
    - Altered mental status
    - Need for mechanical ventilation[TT]
    - Need for extracorporeal membrane oxygenator or right ventricular assist device
  - Aspiration pneumonia with **1 or more** of the following(415)(416):
    - Acute respiratory distress syndrome (eg, PaO_2/FiO_2 ratio of 300 mm Hg or less)
    - Impending or actual respiratory failure
    - Need for invasive or noninvasive mechanical ventilation[TT]
  - *Pneumocystis jirovecii* pneumonia with **1 or more** of the following(417)(418):
    - Hypoxia (eg, PaO_2 60 mm Hg (8.0 kPa) or less despite oxygen therapy)
    - Impending or actual respiratory failure
    - Need for invasive or noninvasive mechanical ventilation[TT]
    - Patient with hematologic malignancy and signs of respiratory deterioration
  - Pneumonia in adult with **1 or more** of the following(419)(420)(421)(422):
    - Need for invasive or noninvasive assisted ventilation[TT]
    - Hemodynamic instability
    - Severe disease, as indicated by **3 or more** of the following:
      - PaO_2/FiO_2 ratio of 250 mm Hg or less
      - Multilobed infiltrates
      - Altered mental status
      - BUN 20 mg/dL (7.1 mmol/L) or greater
      - WBC count less than 4000/mm^3 (4 $x10^9$/L)
      - Platelet count less than 100,000/mm^3 (100 $x10^9$/L)
      - Serum sodium less than 130 mEq/L (mmol/L)
      - Temperature less than 96.8 degrees F (36 degrees C)
  - Pneumonia in child with **1 or more** of the following(423)(424)(425):
    - Impending respiratory failure
    - Need for invasive or noninvasive ventilation[TT]
    - Hemodynamic instability
    - Pulse oximetry less than 92% on greater than 50% inspired oxygen
    - Recurrent apnea
    - Altered mental status
    - Hemolytic uremic syndrome[UU](261)(426)
  - COVID-19 infection with **1 or more** of the following(5)(419)(424)(427)(428)(429)(430):
    - Need for invasive or noninvasive mechanical ventilation
    - Adult with pulse oximetry 90% or lower on 60% inspired oxygen or higher
    - Pregnant patient with pulse oximetry of 94% or lower despite supplemental oxygen
    - Child with pulse oximetry 92% or lower on 50% inspired oxygen or higher

    - Child with recurrent apnea
    - Respiratory acidosis with pH less than 7.20
    - Hemodynamic instability
    - Requirement for frequent hemodynamic measurements
    - Altered mental status
    - Need for intravenous vasopressor or inotrope infusion
    - New end organ dysfunction
- Acute viral illness (eg, influenza, respiratory syncytial virus) with **1 or more** of the following(5)(336)(419)(427)(431):
    - Impending respiratory failure[SS]
    - Need for invasive or noninvasive mechanical ventilation[TT]
    - Hemodynamic instability
    - Requirement for frequent hemodynamic measurements
    - Need for intravenous vasopressor or inotrope infusion
    - Altered mental status that is severe or persistent
    - New end organ dysfunction (eg, Acute kidney injury (stage 2), Acute renal failure (stage 3 acute kidney injury), acute liver failure)
    - Respiratory acidosis with pH less than 7.20
    - Adult with refractory hypoxemia (eg, pulse oximetry 90% or lower on 60% inspired oxygen or higher)
    - Child with pulse oximetry less than 92% on 50% inspired oxygen or higher
    - Child with recurrent apnea
- Pulmonary hypertension with **1 or more** of the following(93)(414)(432)(433):
    - Hemodynamic instability
    - Hypoxemia unresponsive to oxygen supplementation
    - Need for pulmonary artery catheterization (eg, for titration of pulmonary vasodilator agents or monitoring of cardiac output)
    - Need for initiation of parenteral pulmonary vasodilator
    - Need for inhaled nitric oxide or epoprostenol
    - Need for vasopressor or inotropic agent
    - Need for mechanical circulatory support (eg, extracorporeal membrane oxygenator or right ventricular assist device, as bridge to recovery or transplant)
- Obesity hypoventilation with **1 or more** of the following(434)(435):
    - Respiratory distress
    - Need for invasive mechanical ventilation
    - Altered mental status thought to be caused by hypercarbia
- Impending respiratory failure, as indicated by **1 or more** of the following:
    - Child with clinical signs of impending respiratory failure[SS]
    - Clinical signs of respiratory fatigue (eg, use of respiratory accessory muscles, paradoxical motion of the diaphragm,[VV] intercostal retractions)(436)
    - Adult with respiratory rate greater than 30 breaths per minute[A][WW]
    - Arterial partial pressure of oxygen less than 60 mm Hg (8.0 kPa) on 50% oxygen or more or PaO_2/FiO_2 ratio less than 200 mm Hg(415)(437)
    - Arterial partial pressure of carbon dioxide greater than 45 mm Hg (6.0 kPa) with pH less than 7.35
    - Altered mental status due to respiratory disease
- Respiratory failure with **1 or more** of the following(409)(438)(439):
    - New need for acute invasive or noninvasive mechanical ventilation(45)(313)(440)(441)(442)
    - High likelihood of requiring mechanical ventilation within 24 hours
    - Observation in first several hours immediately after extubation from mechanical ventilation
    - Need for close observation and aggressive therapy, such as suctioning, chest physiotherapy, or inhalation treatments at intervals less than 1 hour
    - Pharmacologic ventilatory paralysis

- Venous thromboembolism or deep venous thrombosis with need for systemic or catheter-directed thrombolysis (eg, for limb-threatening or organ-threatening thrombosis, obstructive superior vena cava syndrome)(443)(444)(445)(446)
- Pulmonary embolus (PE) with **1 or more** of the following(447)(448)(449):
    - Hypotension
    - Arterial partial pressure of oxygen less than 60 mm Hg (8.0 kPa) on supplemental oxygen
    - Cardiac arrhythmias of immediate concern
    - Bleeding
    - Need for systemic or catheter-directed thrombolysis
    - High-risk submassive PE (high-intermediate risk), as evidenced by **ALL** of the following:
        - Evidence of right ventricular strain (eg, by echocardiogram or on CT angiogram)
        - Elevated cardiac biomarkers
- Lobectomy or other major thoracic surgery
- Lung transplant(450)
- Need for extracorporeal membrane oxygenation(451)(452)
- Symptomatic upper airway obstruction (eg, laryngeal edema, mass, obstructing foreign body aspiration)(69)(225)(439)(453)(454)(455)
- Massive hemoptysis or pulmonary hemorrhage with **1 or more** of the following(456)(457)(458):
    - Respiratory distress or respiratory failure
    - Airway obstruction (eg, from active bleeding or clot)
    - Hypoxemia unresponsive to oxygen supplementation
    - Hypotension
    - Hemoptysis greater than 200 mL per day
    - Pulmonary vasculitis syndrome with concomitant organ failure requiring ICU care (eg, pulmonary-renal syndrome requiring continuous renal replacement therapy, systemic lupus erythematosus with acute encephalopathy)
- Pertussis and **1 or more** of the following(459):
    - Child younger than 3 months
    - Hemodynamic instability
    - Apnea
    - Respiratory fatigue (eg, loss of whooping inhalation due to fatigue, need for respiratory assistance)
    - Hypoxia (eg, PaO_2 60 mm Hg (8.0 kPa) or less despite oxygen therapy)
    - Impending or actual respiratory failure
- Near drowning (nonfatal submersion) with **1 or more** of the following(63)(64)(67)(460)(461)(462):
    - Need for acute invasive or noninvasive mechanical ventilation
    - Respiratory distress
    - Neurologic impairment requiring monitoring
    - Severe Hypothermia, as indicated by **1 or more** of the following(63)(64):
        - Temperature 89.6 degrees F (32 degrees C) or lower
        - Temperature 95 degrees F (35 degrees C) or lower and severe organ dysfunction, as indicated by **1 or more** of the following:
            - Altered mental status
            - Hemodynamic instability
            - Respiratory depression
            - Pulmonary edema
            - Arrhythmia (eg, Bradycardia, atrial fibrillation, ventricular fibrillation)
            - Acute kidney injury (stage 2) or Acute renal failure (stage 3 acute kidney injury)
            - Other new end organ dysfunction attributable to hypothermia
- Infection or thrombosis of intravenous device with **1 or more** of the following(5)(463)(464):
    - Unstable acute complication (eg, pericardial tamponade, tension pneumothorax, air embolism, active bleeding)

        - Hemodynamic instability
        - Requirement for frequent hemodynamic measurements
        - End organ dysfunction (eg, renal failure, Altered mental status that is severe or persistent)(5)(184)(464)(465)
        - Acute renal complications due to missed dialysis (eg, severe electrolyte abnormalities, uremia, or acidosis)
    - Traumatic rib fracture or fractures with **1 or more** of the following(466)(467)(468):
        - Injury severity score of 19 or greater
        - Forced vital capacity of 25% predicted or less
        - Respiratory insufficiency
        - Flail chest
        - Sternum fracture
        - Vascular injury (eg, heart or great vessels)
    - Pleural effusion with **1 or more** of the following(469):
        - Respiratory insufficiency
        - Hemothorax with active ongoing bleeding
        - Hemodynamic instability
        - Unstable comorbid condition (eg, heart failure)
    - Tension pneumothorax(469)(470)
    - Complications of any thoracic surgery requiring ICU intervention, as indicated by **1 or more** of the following(161):
        - Hemodynamic instability
        - MI with complications (eg, severe arrhythmia, Hypotension)
        - Excessive bleeding or severe coagulopathy
        - Respiratory failure or insufficiency
        - Renal failure
        - Airway instability or obstruction
        - Neurologic deterioration
        - Infection with Hypotension or significant fluid shifts

- **Urology** diagnoses or procedures, including **1 or more** of the following(471)(472):
    - Renal transplant
    - Postobstructive diuresis (eg, following catheterization or stent for obstructive uropathy) greater than 4 L per day requiring large-volume fluid replacement and frequent monitoring and replacement of electrolytes(473)
    - Necrotizing perineal infection (Fournier gangrene)(474)(475)
    - Urinary tract infection with **1 or more** of the following(5)(184)(372)(464)(476)(477):
        - Hemodynamic instability
        - Requirement for frequent hemodynamic measurements (eg, arterial catheter, pulmonary artery catheter)
        - Shock
        - End organ dysfunction (eg, Acute renal failure (stage 3 acute kidney injury), acute respiratory distress syndrome, increase of 2 or more points on Sequential Organ Failure Assessment (SOFA) score)
    - Obstructive pyonephrosis with **1 or more** of the following(5)(184)(464)(476):
        - Hemodynamic instability
        - Requirement for frequent hemodynamic measurements (eg, arterial catheter, pulmonary artery catheter)
        - End organ dysfunction (eg, acute kidney injury, acute respiratory distress syndrome, increase of 2 or more points on SOFA (Sequential Organ Failure Assessment) score)
    - Postoperative complication requiring ICU care, as indicated by **1 or more** of the following(161)(478)(479)(480)(481)(482):
        - Hemodynamic instability
        - Myocardial infarction
        - Excessive bleeding or coagulopathy

        - Respiratory failure or insufficiency
        - Airway instability or obstruction
        - Neurologic deterioration (eg, Altered mental status)
        - Infection with Vital sign abnormality
        - Bowel ischemia(481)
        - Bowel injury with peritonitis
        - Acute renal failure (stage 3 acute kidney injury)
        - Severe electrolyte abnormalities (eg, hyponatremia with Altered mental status, or monitoring need requires ICU)
    - Preoperative or postoperative patient requiring ICU intervention, such as pulmonary artery monitoring, mechanical ventilation, or extensive nursing care(161)

Intensive Care Discharge Guidelines

- ICU discharge appropriate, as indicated by **1 or more** of the following[XX](1)(5)(12)(108)(161):
    - Physiologic status has stabilized such that ICU care is no longer needed, as indicated by **ALL** of the following:
        - No ICU-specific intervention or monitoring currently needed or anticipated
        - Discontinuation of parenteral vasopressor, inotropic, antihypertensive, antianginal, and antiarrhythmic agents for more than 4 hours with adequate stability (eg, mean arterial pressure (MAP) greater than 65 mm Hg and less than 110 mm Hg MAP Calculator, no arrhythmia or hemodynamically stable intermittent arrhythmia)[H](5)(108)
        - Patient airway is patent (native airway or tracheostomy) and airway clearance is adequate for next level of care, as indicated by **1 or more** of the following(108):
            - Adequate cough
            - Tracheal suctioning feasible at receiving unit or home
        - No ICU-assisted ventilation[F] currently needed or anticipated, as indicated by **1 or more** of the following:
            - Patient did not need assisted ventilation.[F]
            - Extubation and discontinuation of assisted ventilation[F] with stable respiratory status for at least 4 hours
            - Stable chronic ventilator or long-term weaning needs
        - No significant active bleeding(108)(161)
        - No evidence of active ischemia(161)
        - No evidence of end organ hypoperfusion (eg, lactate level less than 2.5 mmol/L (22.5 mg/dL))(161)
        - Neurologic checks not required, or required less frequently than every 4 hours(161)
        - No requirement for ICU level of nursing care (ie, patient without need for nursing ratio of 1 or 2 patients per nurse)
    - Physiologic status has deteriorated such that active interventions requiring ICU level of care are no longer planned.(108)(483)(484)

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Footnotes

[A] Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture. [A in Context Link 1, 2]

[B] The mean arterial pressure takes into account both systolic and diastolic blood pressure readings.(9) [B in Context Link 1, 2]

[C] Pulse oximetry (SpO_2) may underestimate arterial saturation (SaO_2), especially in people with dark skin pigmentation, thick skin, cold body temperature, or who are wearing colored nail polish, and thus may not accurately detect hypoxemia.(27) Where appropriate, pulse oximetry should be measured after warming fingers or removing nail polish. Pulse oximetry results should be assessed within the context of the patient's clinical presentation, and performance of an arterial blood gas considered for a more accurate assessment of oxygenation if indicated.(27) Specific target values may require adjustment when care is delivered at high altitude.(28) [C in Context Link 1, 2, 3]

[D] An ionized calcium level may be more accurate than a total serum calcium level, especially in the setting of acidosis, renal failure, multiple transfusions, hypoalbuminemia, or the presence of a paraprotein.(30)(31) [D in Context Link 1, 2]

[E] A dedicated burn unit is recommended.(33)(34) [E in Context Link 1]

[F] Ventilatory assistance includes noninvasive methods (eg, high-flow nasal cannula (HFNC), CPAP, BPAP) and invasive methods (intubation with mechanical ventilation). [F in Context Link 1, 2, 3, 4]

[G] Many indications for pulmonary artery catheterization (right-sided heart catheterization) do not require ongoing hemodynamic monitoring and can be performed as an outpatient in a catheterization laboratory; examples include evaluation of suspected intracardiac shunt or pulmonary hypertension, determination of acute response to pulmonary vasodilator therapy, post-heart transplant patient follow-up, and endomyocardial biopsy.(90) [G in Context Link 1]

[H] Low doses of vasopressor or inotropic agents may be administered in some intermediate care units depending on unit capabilities, staffing, and policies.(108) [H in Context Link 1, 2]

[I] Cerebral hyperperfusion syndrome manifests as an increase in cerebral blood flow following a revascularization procedure such as carotid stenting, endarterectomy, or thrombectomy. Symptoms may include new-onset headache, facial and eye pain, and lethargy; severe cases may present with seizure, neurologic deficits, intracerebral hemorrhage, cerebral edema, or coma. Patients are usually hypertensive. The diagnosis is made by evidence of increased cerebral blood flow on transcranial duplex scan and lack of evidence for postoperative carotid occlusion or new ischemic event on brain imaging studies.(127) [I in Context Link 1]

[J] Adrenal crisis presents as life-threatening multiorgan dysfunction or failure with shock. It may occur as the result of acute adrenal insufficiency that is primary (eg, autoimmune destruction, infection, hemorrhage, mass of the adrenal gland), secondary adrenal insufficiency (eg, mass, hemorrhage, or infiltrative disorder of the pituitary gland), or tertiary (eg, mass or infiltrative disorder of the hypothalamus), with the acute crisis often triggered by stress (eg, infection or hemorrhage).(131)(132) Iatrogenic causes include acute cortisol deficiency following glucocorticosteroid cessation or taper, radiation therapy, and drugs, including immune checkpoint inhibitors.(131)(132) [J in Context Link 1]

[K] Serum osmolality can be directly measured in the serum or estimated. Various equations have been proposed to estimate serum osmolality, a common one being (sodium ion (in mEq/L) x 2) + (glucose (in mg/dL)/18) + (blood urea nitrogen (BUN, in mg/dL)/2.8). A calculator for serum osmolality is available at www.mdcalc.com/calc/91/serum-osmolality-osmolarity. [K in Context Link 1]

[L] The Burch-Wartofsky Point Scale assigns points for thermoregulatory function, central nervous system effects, gastrointestinal or hepatic dysfunction, cardiovascular symptoms, and precipitating events.(149)(151) An online calculator is available at www.mdcalc.com/burch-wartofsky-point-scale-bwps-thyrotoxicosis. [L in Context Link 1]

[M] Patients with cardiogenic shock due to thyrotoxic storm may require cautious use of short-acting intravenous beta-blockers, intravenous vasopressor agents, a ventricular assist device for hemodynamic support, and therapeutic plasma exchange to decrease thyroid hormone levels.(153) [M in Context Link 1]

[N] Pituitary apoplexy is caused by sudden hemorrhage or infarction which can occur in a healthy pituitary gland (eg, triggered by hypertensive crisis, trauma, or hypotensive-induced ischemia), in a patient with a pituitary mass (eg, adenoma, pituitary neuroendocrine tumor, metastasis), pituitary inflammation (eg, hypophysitis, tuberculosis), or after massive postpartum hemorrhage (Sheehan syndrome).(157) Presenting symptoms include headache, visual disturbance or blindness, altered mental status, focal cranial nerve deficits, and hemodynamic instability, and it is best diagnosed by MRI.(129)(157)(158) Acute management includes stress dose corticosteroids, correction of hyponatremia, cabergoline if prolactinoma is present, thyroid hormone replacement if needed, and possible emergent surgical pituitary tumor resection.(158) [N in Context Link 1]

[O] A calculator is available at www.mdcalc.com/apache-ii-score. [O in Context Link 1]

[P] Seriously injured patients may require transfer to a level 1 or level 2 trauma center.(201) [P in Context Link 1]

[Q] Pancreaticoduodenectomy (also known as Whipple procedure) is a complex operation involving removal of the head of the pancreas, portions of the duodenum, gallbladder, and (sometimes) the stomach antrum, with anastomosis of the stomach to the jejunum, and reimplantation of the remaining body and tail of the pancreas and the common bile duct into a blind distal "Roux-en-Y" limb of the jejunum.(196) Although the mortality of this procedure has greatly improved since its introduction, especially in high-volume centers, the procedure is still characterized by high morbidity.(196) [Q in Context Link 1]

[R] Patients with sickle cell disease may have falsely elevated pulse oximetry due to higher levels of methemoglobin and carboxyhemoglobin that are generated from the hemolysis that occurs with active sickling.(244) [R in Context Link 1]

[S] Disseminated intravascular coagulation (DIC) is diagnosed based on the International Society on Thrombosis and Hemostasis score, which includes factors for platelet count, D-dimer levels, prothrombin time, and fibrinogen. A score of 5 or greater, in association with any clinical disorder known to cause DIC, is indicative of DIC.(248)(249) Clinical disorders associated with DIC include sepsis, severe pancreatitis, severe tissue injury, retained placenta or placenta abruption, acute fatty liver of pregnancy, certain cancers, venomous snake bites, prostate surgery complications, and damage to a blood vessel wall (eg, aortic aneurysm, cavernous hemangioma).(248)(250) A calculator is available at www.mdcalc.com/calc/10203/isth-criteria-disseminated-intravascular-coagulation-dic. The sepsis-induced coagulation (SIC) score, which includes criteria for INR, platelet count, and SOFA score, can detect earlier stages of DIC in patients with sepsis.(249) A calculator for the SIC score is available at www.mdcalc.com/calc/10301/sepsis-induced-coagulopathy-sic-score. [S in Context Link 1]

[T] Although hyperleukocytosis is defined by a white blood cell count greater than 100,000/mm^3 (100 $x10^9$/L), leukostasis refers to end organ damage associated with an elevated white blood cell count. No single white blood cell count threshold exists for leukostasis to occur as different types of leukemia cells cause leukostasis at widely variable cell counts (in part due to the different sizes of leukemic blast cells).(239) Leukostasis may occur at white blood cell counts even less than 100,000/mm^3 (100 $x10^9$/L) in certain forms of acute myeloid leukemia, but typically occurs at much higher white blood cell counts (eg, greater than 400,000/mm^3 (400 $x10^9$/L)) in acute lymphoid leukemia.(234) [T in Context Link 1]

[U] Hemophagocytic lymphohistiocytosis is a hyper-immune response that can be primary (due to congenital deficiencies in cytolytic pathway proteins), or secondary to triggering conditions including malignancy, stem cell transplant, CAR T-cell therapy, chronic infection, and autoimmune diseases.(254) It often presents with high fever, enlarged lymph nodes, hepatosplenomegaly, elevated ferritin, and pancytopenias, and it can rapidly progress to multiorgan failure and death if untreated.(254) [U in Context Link 1]

[V] Deep neck infections may spread through local fascial planes and cause airway obstruction, mediastinitis, cavernous sinus thrombosis, and necrotizing pneumonia and empyema, and may cause sepsis by hematogenous spread.(215)(225) [V in Context Link 1]

[W] The Centers for Disease Control and Prevention (CDC) case definitions for toxic shock are available at https://ndc.services.cdc.gov/case-definitions/toxic-shock-syndrome-2011/. [W in Context Link 1]

[X] Status epilepticus is defined as continuous clinical or electrical seizure activity or repetitive seizures with incomplete recovery for a period of at least 5 minutes.(286)(287) [X in Context Link 1]

[Y] Patients should be in a monitored setting when receiving antivenin due to the high risk for anaphylactic reactions.(289) [Y in Context Link 1]

[Z] Continuous renal replacement therapy is recommended for patients who are hemodynamically unstable, or have cerebral edema or acute brain injury.(57)(294) [Z in Context Link 1]

[AA] The modified Erasmus GBS respiratory insufficiency scale predicts the likelihood of respiratory insufficiency and need for mechanical ventilation based on factors for days between onset and hospitalization, the presence or absence of facial or bulbar weakness, and the sum Medical Research Council (MRC) strength scores of 6 muscle groups. A calculator can be found at https://qxmd.com/calculate/calculator_527/erasmus-gbs-respiratory-insufficiency-score-egris#.(316) [AA in Context Link 1]

[BB] The decision to intubate a patient with myasthenia gravis is a clinical one based on signs of respiratory distress, paradoxical breathing pattern, tachypnea, declining lung volumes and respiratory muscle strength on bedside testing, and difficulty protecting the airway. Parameters that may help guide this clinical decision include decreasing vital capacity (eg, less than 20 mL/kg of ideal body weight), a negative inspiratory force measurement less than -30 cm H_2O (-2942 Pa), or a positive expiratory force measurement greater than 40 cm H_2O (3923 Pa).(319) [BB in Context Link 1]

[CC] Respiratory isolation for 24 hours may be needed for patients with suspected meningococcal infection.(320)(321) [CC in Context Link 1]

[DD] If available, a neurocritical care or stroke unit that incorporates rehabilitation is preferred.(328)(329) [DD in Context Link 1]

[EE] Patients receiving thrombolytic agents may be transferred out of the ICU after 24 hours if there is no evidence of hemorrhagic conversion on examination and 24-hour post-thrombolysis brain imaging study and no other indications for ICU monitoring.(328) [EE in Context Link 1]

[FF] Patients with acute spinal cord injury may experience hemodynamic instability and may require vasoactive agents to optimize spinal cord perfusion.(334) Patients with cervical spine injury may have impaired diaphragm function with resultant respiratory insufficiency that may require mechanical ventilation.(334) [FF in Context Link 1]

[GG] Acute flaccid paralysis is historically caused by myelitis due to poliovirus but has more recently been associated with some enterovirus and coxsackie virus strains.(336) Affecting the anterior spinal horn cells, it presents with rapid onset of limb weakness and can cause respiratory failure in about 1/3 of patients.(335)(336) Other manifestations include acute mental status change and autonomic dysfunction including labile blood pressure and arrhythmia.(336) [GG in Context Link 1]

[HH] If available, a neurocritical care unit with neurosurgical capabilities is preferred.(329)(337) [HH in Context Link 1]

[II] A dedicated neurocritical care unit is recommended.(343) [II in Context Link 1]

[JJ] Status dystonicus is a life-threatening movement disorder that can either occur in patients with pre-existing dystonic conditions (eg, cerebral palsy, monogenic dystonia) with precipitation by a triggering event (eg, acute illness, fever, or dehydration), or can present as a first manifestation of a genetic dystonia, metabolic disorder (eg, Lesch-Nyhan syndrome), or an infection or inflammatory disorder affecting the central nervous system. Status dystonicus may cause hyperpyrexia, rhabdomyolysis, acute kidney injury, aspiration, severe pain, and respiratory failure. ICU monitoring is recommended for respiratory monitoring with potential need for intubation due to both the dystonia and the need for high doses of respiratory depressant medications.(74)(355) [JJ in Context Link 1]

[KK] Initial treatment is systemic anticoagulation with close observation for signs of progression, seizures, and elevated intracranial pressure.(356)(357) Patients with clot propagation despite anticoagulation may require endovascular intervention.(356) Decompressive craniectomy may be needed for impending herniation.(356) Septic cerebral venous thrombosis (most commonly cavernous sinus thrombosis) is a type of cerebral venous thrombosis that is associated with local infection (eg, sinusitis or mastoiditis). Treatment requires both anticoagulation and antibiotics.(357) [KK in Context Link 1]

[LL] Example diagnoses include the Marburg, tumefactive, and Balo variants of multiple sclerosis; lupus encephalitis; limbic encephalitis; anti-NMDA receptor encephalitis; steroid-responsive encephalopathy with autoimmune thyroiditis; acute disseminated encephalomyelitis; and neuromyelitis optica spectrum disorder.(359) [LL in Context Link 1]

[MM] Complications of pregnancy associated with disseminated intravascular coagulation include puerperal sepsis, peripartum hemorrhage, placenta abruption, retained placenta, retained stillbirth, anaphylactoid syndrome of pregnancy (amniotic fluid embolism), acute fatty liver of pregnancy, preeclampsia, and eclampsia.(250) [MM in Context Link 1]

[NN] Symptoms of mild to moderate ovarian hyperstimulation syndrome include abdominal distention, mild dyspnea, enlarged ovaries, gastrointestinal symptoms, and evidence of mild hemoconcentration. More severe stages can be a serious life-threatening condition presenting with hypotension, pericardial effusion, massive hydrothorax, severe dyspnea, acute respiratory

distress syndrome, severe electrolyte imbalance, renal impairment, and hypercoagulable state. It is seen with assisted reproduction following ovarian stimulation with gonadotropin hormones.(383) [NN in Context Link 1]

[OO] A score of less than 6 on the Sepsis in Obstetric Score has been shown to be helpful in identifying pregnant patients who do not require ICU hospitalization, with a negative predictive value of 98.6%. An online calculator is available at https://perinatology.com/calculators/Sepsis%20Calculator.htm.(385) [OO in Context Link 1]

[PP] Initial treatment is systemic anticoagulation with close observation for signs of progression, seizures, and elevated intracranial pressure.(356)(357) Patients with clot propagation despite anticoagulation may require endovascular intervention.(356) Decompressive craniectomy may be needed for impending herniation.(356) Septic cerebral venous thrombosis (most commonly cavernous sinus thrombosis) is a subtype of cerebral venous thrombosis that is associated with local infection (eg, sinusitis or mastoiditis). Treatment requires both anticoagulation and antibiotics.(357) [PP in Context Link 1]

[QQ] Patients with pulmonary arterial hypertension may require monitoring with arterial and pulmonary artery catheterization during induction and delivery, and for 72 hours following delivery, due to risk of hemodynamic instability caused by fluid shifts during that period.(388) [QQ in Context Link 1]

[RR] Based on expert opinion and observational studies, a specialty society guideline recommends intravenous thrombolysis within 4.5 hours of symptom onset for patients with acute central retinal artery occlusion who have no contraindications to thrombolytic therapy.(390) The same guideline recommends catheter-directed thrombolysis within 6 hours of symptom onset for patients with acute central retinal artery occlusion who are not candidates for IV thrombolysis (eg, because of recent surgery, gastrointestinal hemorrhage, or other significant bleeding risk; iatrogenic bleeding risk is presumed to be lower than IV administration since much lower thrombolytic doses can be used).(390) Guidelines for acute central retinal artery occlusion do not address the length of time that ICU monitoring is needed following thrombolytic therapy.(390) However, for other types of stroke, a specialty society guideline recommends that patients receiving thrombolytic agents may be transferred out of the ICU after 24 hours if there is no evidence of hemorrhagic conversion and no other indications for ICU monitoring.(328) [RR in Context Link 1]

[SS] Factors included in the subjective assessment of impending respiratory failure in children include the severity of tachypnea and tachycardia; presence of chest retractions, central cyanosis, Altered mental status, and difficulty feeding due to respiratory distress.(406) [SS in Context Link 1, 2, 3, 4]

[TT] Continuous noninvasive ventilation may be administered in some intermediate care units depending on unit capabilities, staffing, and policies.(44) [TT in Context Link 1, 2, 3, 4, 5, 6, 7]

[UU] Hemolytic uremic syndrome (HUS) is a thrombotic microangiopathy characterized by hemolytic anemia, thrombocytopenia, and acute kidney injury. While usually associated with gastroenteritis from Shiga toxin and Shiga-toxin-producing bacteria (eg, *Escherichia coli*, *Shigella*, and *Citrobacter* species), invasive *Streptococcus pneumoniae* infection is the most common nondiarrheal etiology and is responsible for about 5% of cases of HUS.(261)(426) Severe cases may involve other organs, including the central nervous system (eg, encephalopathy, seizures), pancreas (pancreatitis, diabetes), liver (hepatitis, jaundice), and heart (eg, hypertension, heart failure).(261)(426) [UU in Context Link 1]

[VV] Paradoxical diaphragm motion or abdominal paradox occurs when the diaphragm is severely fatigued or paralyzed and is pulled up into the chest cavity during inspiration, opposite its normal physiologic direction.(436) [VV in Context Link 1]

[WW] Interpretation of respiratory rate requires clinical judgment and consideration of several patient-specific factors, such as the patient's baseline respiratory rate and whether the respiratory rate is indicative of significant underlying respiratory or other pathology. In addition, the measurement of the respiratory rate is subject to error. A medical textbook states that because there can be significant breath-to-breath variation in respiratory rate, the rate should be assessed for at least 30 seconds, preferably for a full minute.(17) This same textbook also states that new technologies designed to measure the respiratory rate have not proved to be clinically useful.(17) [WW in Context Link 1]

[XX] The intensive care guideline should not be used independently but in conjunction with the appropriate Inpatient & Surgical Care or General Recovery Care guideline. [XX in Context Link 1]

Definitions

Acute kidney injury (stage 2)

- Acute kidney injury (stage 2) with significant clinical findings, as indicated by **ALL** of the following(1)(2)(3):
  - Acute[A] kidney injury (stage 2), as indicated by **1 or more** of the following[B]:
    - Acute[A] rise in serum creatinine between 2 and 2.9 times its baseline value[C] (increase of 3 times baseline would qualify as stage 3 acute kidney injury)

- Decreased urine output,[D] as indicated by **ALL** of the following:
  - Assessment of urine output appropriate (eg, adequate volume status, no urinary obstruction)
  - Urine output less than 0.5 mL/kg/hr for 12 hours
- Significant clinical findings, as indicated by **1 or more** of the following[B]:
  - Hemodynamic instability
  - Worsening clinical status despite observation care (eg, rising creatinine or urine output remains less than 0.5 mL/kg/hr)
  - Altered mental status
  - Cardiac arrhythmias of immediate concern
  - Severe hypertension (SBP greater than 180 mm Hg or DBP greater than 120 mm Hg in adults or greater than the 95th percentile for age, gender, and height plus 30 mm Hg in pediatric patients)(4)(5)
  - Significant respiratory findings (eg, pulmonary edema) that are severe (eg, Hypoxemia) or persist despite observation care
  - Clinically significant electrolyte abnormality that is severe (eg, hyperkalemia with severe ECG findings)[E] or persists despite observation care
  - Clinically significant metabolic abnormality (eg, metabolic acidosis) that is severe or persists despite observation care

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Footnotes

A. A specialty society practice guideline defines acute kidney injury as a change in renal function that is known or presumed to have occurred over the previous 7 days.(1)
B. In determining the stage of acute kidney injury (eg, stage 2 vs stage 3), patients should be staged according to the criteria that give them the highest stage.(1)
C. If the baseline serum creatinine is not known, absent known or suspected chronic kidney disease, a baseline serum creatinine can be approximated by assuming an estimated GFR of 75 mL/min/1.73m^2 (1.25 mL/sec/1.73m^2).(1)
D. The use of urine output as an indicator of acute kidney injury is important, as change in urine output can occur before a noticeable rise in creatinine.(1)(3) It is also suggested that urine output can identify patients with acute kidney injury that may otherwise be missed.(1)(3) However, urine output as a metric is only valid under the appropriate conditions. For example, the patient must not be hypovolemic, there must not be urinary obstruction, and the collection and measurement of urinary output must be complete and accurate.(1)(3) Use of urinary output to define acute kidney injury can be inaccurate in obese patients due to the nonlinear relationship between body weight and urine output. For example, in a 100-kg (220-lb) patient, urine output of 45 mL/hr over 12 hours (adequate output) would qualify as stage 2 acute kidney injury.(1)(3)
E. ECG findings include peaked T-waves, PR interval greater than 0.20 seconds, QRS interval greater than 0.12 seconds, and arrhythmia.(6)(7)

Acute renal failure (stage 3 acute kidney injury)

- Acute[A] kidney injury (stage 3),[B] as indicated by **1 or more** of the following[C](1)(2)(3):
  - Acute[A] rise in serum creatinine to 3 times its baseline value[D] or higher
  - Acute[A] rise in serum creatinine to at least 1.5 times its baseline value[D] (increase of 50%) and serum creatinine is greater than 4 mg/dL (354 micromoles/L)
  - In child younger than 18 years, estimated GFR less than 35 mL/min/1.73m^2 (0.59 mL/sec/1.73m^2) eGFR - Pediatric Calculator and **1 or more** of the following:
    - Acute[A] rise in serum creatinine to at least 1.5 times its baseline value[D] (increase of 50%)
    - Within past 48 hours, rise in serum creatinine greater than 0.3 mg/dL (26.5 micromoles/L)
  - Decreased urine output,[E] as indicated by **ALL** of the following:
    - Assessment of urine output appropriate (eg, adequate volume status, no urinary obstruction)
    - Oligoanuria, as indicated by **1 or more** of the following:
      - Urine output less than 0.3 mL/kg/hr for 24 hours
      - Anuria (urine output less than 0.1 mL/kg/hr) for 12 hours
  - Initiation of renal replacement therapy required(4)

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Footnotes

A. A specialty society practice guideline defines acute kidney injury as a change in renal function that is known or presumed to have occurred over the previous 7 days.(1)
B. For purposes of these criteria, the term acute renal failure can be viewed as stage 3 acute kidney injury in the KDIGO classification system.(1) Of note, in the ICD-10 diagnostic coding system (main coding scheme used in labeling diagnoses), the relevant codes covering the clinical entity of significant acute kidney injury use the term acute kidney failure (eg, code N17.9).
C. In determining the stage of acute kidney injury (eg, stage 2 vs stage 3), patients should be staged according to the criteria that give them the highest stage.(1)
D. If the baseline serum creatinine is not known, absent known or suspected chronic kidney disease, a baseline serum creatinine can be approximated by assuming an estimated GFR of 75 mL/min/1.73m^2 (1.25 mL/sec/1.73m^2).(1)
E. The use of urine output as an indicator of acute kidney injury is important, as change in urine output can occur before a noticeable rise in creatinine.(1)(3) It is also suggested that urine output can identify patients with acute kidney injury that may otherwise be missed.(1)(3) However, urine output as a metric is only valid under the appropriate conditions. For example, the patient must not be hypovolemic, there must not be urinary obstruction, and the collection and measurement of urinary output must be complete and accurate.(1)(3) Use of urinary output to define acute kidney injury can be inaccurate in obese patients due to the nonlinear relationship between body weight and urine output. For example, in a 100-kg (220-lb) patient, urine output of 45 mL/hr over 12 hours (adequate output) would qualify as stage 2 acute kidney injury.(1)(3)

Altered mental status

- Altered mental status (ie, different from baseline), as indicated by **1 or more** of the following(1)(2)(3)(4):
  - Confusional state (eg, disorientation, difficulty following commands, deficit in attention)
  - Lethargy (eg, awake or arousable, but with drowsiness; reduced awareness of self and environment)
  - Obtundation (ie, arousable only with strong stimuli, lessened interest in environment, slowed responses to stimulation)
  - Stupor (ie, may be arousable but patient does not return to normal baseline level of awareness)

- Coma (ie, not arousable)

References

1. Maher PJ. Confusion. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Elsevier; 2023:126-131.e1.
2. Lei C, Smith C. Depressed consciousness and coma. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Elsevier; 2023:117-125.e1.
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Altered mental status that is severe or persistent

- Altered mental status (ie, different from baseline) that is severe or persistent, as indicated by **1 or more** of the following(1)(2)(3)(4):
  - Confusional state (eg, disorientation, difficulty following commands, deficit in attention) that persists despite appropriate treatment (eg, of underlying cause, despite observation care)
  - Lethargy (eg, awake or arousable, but with drowsiness; reduced awareness of self and environment) that persists despite appropriate treatment (eg, of underlying cause, despite observation care)
  - Obtundation (ie, arousable only with strong stimuli, lessened interest in environment, slowed responses to stimulation)
  - Stupor (ie, may be arousable but patient does not return to normal baseline level of awareness)
  - Coma (ie, not arousable)

References

1. Maher PJ. Confusion. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Elsevier; 2023:126-131.e1.
2. Lei C, Smith C. Depressed consciousness and coma. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Elsevier; 2023:117-125.e1.
3. Abraham G, Maher PJ. Delirium and dementia. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Elsevier; 2023:1269-1280.e2.
4. Holler-Managan YF. Neurologic evaluation. In: Kliegman RM, et al., editors. Nelson Textbook of Pediatrics. 22nd ed. Elsevier; 2025:3550-3560.

American Society of Anesthesiologists (ASA) Physical Status Classification System

- American Society of Anesthesiologists (ASA) Physical Status Classification System[A](2)(3)(4)(5)(6)(7)
  - Class I: Normal healthy patient
  - Class II: Patient with mild systemic disease
  - Class III: Patient with severe systemic disease
  - Class IV: Patient with severe systemic disease that is constant threat to life
  - Class V: Moribund patient who is not expected to survive without the operation

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Footnotes

A. Note: There is no accepted standardization of terms used in this classification system.(1)

Bradycardia

- Bradycardia[A], as indicated by **1 or more** of the following(1)(2)(3):
  - Heart rate less than 50 beats per minute in adult (age 18 years or older)[B]
  - Heart rate less than 55 beats per minute in adolescent 13 to 17 years of age
  - Heart rate less than 60 beats per minute in child or adolescent 6 to 12 years of age
  - Heart rate less than 65 beats per minute in child 3 to 5 years of age
  - Heart rate less than 70 beats per minute in child 1 to 2 years of age
  - Heart rate less than 80 beats per minute in infant 6 to 11 months of age
  - Heart rate less than 90 beats per minute in infant 3 to 5 months of age

References

1. Kusumoto FM, et al. 2018 ACC/AHA/HRS guideline on the evaluation and management of patients with bradycardia and cardiac conduction delay: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines and the Heart Rhythm Society. Circulation 2019;140(8):e382-e482. DOI: 10.1161/CIR.0000000000000628. (Reaffirmed 2024 Aug)
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3. Anderson CC, Kapoor S, Mark TE. Pediatric parameters and equipment. In: Anderson CC, Kapoor S, Mark TE, editors. The Harriet Lane Handbook: A Manual for Pediatric House Officers. 23rd ed. Elsevier; 2024:i-iii.

Footnotes

A. Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

B. An evidence-based specialty society guideline has defined bradycardia in adults as less than 50 beats per minute.(1)

Cardiac arrhythmias of immediate concern

- Cardiac arrhythmias or findings of immediate concern, as indicated by **1 or more** of the following(1)(2)(3):
  - Heart rhythms that are inherently dangerous or unstable, as indicated by **1 or more** of the following(4)(5)(6):
    - Resuscitated ventricular fibrillation or cardiac arrest
    - Ventricular escape rhythm
    - Sustained ventricular tachycardia (30 seconds or more of ventricular rhythm at greater than 100 beats per minute)
    - Nonsustained ventricular tachycardia and **1 or more** of the following:
      - Suspected cardiac ischemia as cause or consequence of ventricular tachycardia
      - Acute myocarditis
  - Unstable cardiac conduction defects, as indicated by **1 or more** of the following(5)(7):
    - Type II second-degree atrioventricular block
    - Third-degree atrioventricular block

    - New-onset left bundle branch block with suspected myocardial ischemia
- Any heart rhythm and **1 or more** of the following(4)(8)(9)(10):
    - Continuous long-term ECG monitoring needed (eg, initiation of drug requiring monitoring beyond observation care)
    - Patient has automatic implanted cardioverter-defibrillator that is repeatedly firing, malfunctioning, or in need of immediate adjustment of settings beyond scope of observation care.
- Heart rhythms of concern due to **1 or more** of the following:
    - Hypotension
    - Respiratory distress
    - Association with other significant symptoms (eg, syncope)(8)(9)

References

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Dehydration that is severe or persistent

- Dehydration that is severe or persistent, as indicated by **1 or more** of the following(1)(2)(3)(4):
    - Clinical findings of severe dehydration, as indicated by **1 or more** of the following:
        - Hemodynamic instability
        - Acute renal failure (stage 3 acute kidney injury)
        - Acute kidney injury (stage 2)
        - Serum sodium greater than 150 mEq/L (mmol/L)
    - Dehydration that is persistent, as indicated by **ALL** of the following:
        - Oral rehydration therapy not tolerated or insufficient to adequately correct dehydration
        - Dehydration persists despite appropriate treatment (eg, intravenous fluids, observation care)

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Hemodynamic instability

- Hemodynamic instability, as indicated by **1 or more** of the following(1)(2)(3)(4)(5)(6):
  - Vital sign abnormality not corrected by appropriate treatment, as indicated by **1 or more** of the following[A]:
    - Tachycardia that persists despite appropriate treatment (eg, treatment of underlying cause, despite observation care)
    - Hypotension that persists despite appropriate treatment (eg, treatment of underlying cause, despite observation care)
    - Orthostatic hypotension that persists despite appropriate treatment (eg, treatment of underlying cause, despite observation care)
  - Hypotension that is severe, as indicated by **ALL** of the following:
    - Hypotension
    - Severe hypotension, as indicated by **1 or more** of the following:
      - Lactate of 2.0 mmol/L (18 mg/dL) or more secondary to hypotension (ie, hypoperfusion)[B](3)(6)
      - Metabolic acidosis (arterial or venous[C] pH less than 7.35) not otherwise explained
      - Mean arterial pressure[A][D] less than 65 mm Hg MAP Calculator
      - IV inotropic or vasopressor medication required to maintain adequate blood pressure or perfusion

References

1. Puskarich MA, Jones AE. Shock. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Elsevier; 2023:34-41.e1.
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Footnotes

A. Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

B. There are numerous causes of an elevated lactate level. The most common are cardiogenic or hypovolemic shock, severe heart failure, severe trauma, or sepsis. However, there are also other etiologies to consider such as vigorous exercise, seizures, liver disease, or medication use (eg, metformin, beta2-agonists); therefore, interpretation of elevated lactate levels requires consideration of the clinical context (eg, well-appearing post exercise vs hypotensive). The severity and persistence of the elevation can sometimes be helpful in this differentiation. In most instances of lactic acidosis, blood pH is less than 7.35, with a serum bicarbonate level 20 mEq/L (mmol/L) or lower. However, a coexisting respiratory or metabolic alkalosis can mask these findings.(6)(7)(8)(9)(10)

C. Analyses have concluded that venous and arterial pH measurements are highly correlated, with small differences between them that are usually not clinically significant.(11)(12)(13)(14)(15) If a mixed acid-base disorder is suspected, an arterial blood gas is indicated.(12)

D. The mean arterial pressure (MAP) takes into account both SBP and DBP readings.(1)(4)

Hypotension

- Hypotension, as indicated by **ALL** of the following(1)(2)(3)(4)(5):
  - Not patient baseline (eg, healthy adult with low SBP) or intentional therapeutic goal (eg, low SBP as treatment goal in heart failure)
  - Low blood pressure, as indicated by **1 or more** of the following:
    - SBP less than 90 mm Hg in adult or child age 10 years or older[A][B]
    - Mean arterial pressure[C] less than 70 mm Hg in adult or child age 10 years or older[A][B] MAP Calculator
    - Decrease in baseline mean arterial pressure of 25% or more,[C] with significant signs or symptoms due to lower blood pressure (eg, near syncope, syncope, chest pain)[A][B] MAP Calculator
    - SBP less than sum of 70 mm Hg plus twice patient's age in years in child 1 to 9 years of age[A][B]
    - SBP less than 70 mm Hg in infant 1 to 11 months of age[A][B]

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Footnotes

A. Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

B. Interpretation of blood pressure requires clinical judgment and consideration of several patient-specific factors, such as the patient's baseline blood pressure, medications, and clinical impact. For example, an elderly patient with heart failure may be prescribed medication with the intentional goal of a reduced pressure. If the patient's baseline SBP is 92 mm Hg to 96 mm Hg, absent signs or symptoms of hypotension, an emergency department SBP of 86 mm Hg should not automatically be categorized as hypotensive, even though this is below the cutoff of 90

mm Hg. This would hold for mean arterial pressure (MAP) as well. If the baseline blood pressure is 92 mm Hg systolic and 64 mm Hg diastolic (MAP 73.3 mm Hg), absent signs or symptoms of hypotension, an emergency department blood pressure of 86 mm Hg systolic and 58 mm Hg diastolic (MAP 67.3 mm Hg) may not denote hypotension, even though the MAP is below 70 mm Hg. Conversely, a patient with a baseline blood pressure of 150 mm Hg systolic and 90 mm Hg diastolic (MAP 110 mm Hg), with signs or symptoms of hypotension, can be considered hypotensive even with an emergency department blood pressure of 100 mm Hg systolic and 80 mm Hg diastolic (MAP 86.7 mm Hg). The numeric values included in this definition are provided to allow for consistency in terms of a technical definition of the term hypotension. Whether a blood pressure below the technical threshold is clinically meaningful is a matter of persistence, context, and clinical judgment.

C. The mean arterial pressure (MAP) takes into account both SBP and DBP readings.

Hypothermia

- Hypothermia indicated by core (eg, rectal) body temperature less than or equal to 95 degrees F (35 degrees C)(1)

References

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Hypoxemia

- Hypoxemia, as indicated by **1 or more** of the following(1):
  - Patient without baseline need for supplemental oxygen with oxygen saturation (SpO_2) of less than 90% or arterial blood gas partial pressure of oxygen (PaO_2) of less than 60 mm Hg (8.0 kPa) on room air[A][B]
  - Patient without baseline need for supplemental oxygen who now requires supplemental oxygen to keep SpO_2 greater than 89% or PaO_2 greater than 59 mm Hg (7.9 kPa)[A]
  - Patient with baseline need for supplemental oxygen who now requires increased supplemental oxygen to maintain oxygenation at baseline or acceptable level

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Footnotes

A. Pulse oximetry (SpO_2) may underestimate arterial saturation (SaO_2), especially in people with dark skin pigmentation, thick skin, cold body temperature, or who are wearing colored nail polish, and thus may not accurately detect hypoxemia.(2) Where appropriate, pulse oximetry should be measured after warming fingers or removing nail polish. Pulse oximetry results should be assessed within the context of the patient's clinical presentation, and performance of an arterial blood gas considered for a more accurate assessment of oxygenation if indicated.(2) Specific target values may require adjustment when care is delivered at high altitude.(3)

B. Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

Immunosuppressed

- Immunosuppressed refers to a degree of immunocompromise that is clinically significant (eg, affects choice of antimicrobial agent due to possibility of atypical organism, necessitates additional isolation, or leads to clinically relevant increase in likelihood of severe infection)[A]; examples include(1)(2)(3)(4):
  - Neutropenia (absolute neutrophil count less than 500/mm^3 (0.5 x10^9/L), or expected to decrease to this level within the next 48 hours)
  - Drug-induced immunosuppression (ie, current usage); examples include:
    - Systemic corticosteroids[B](6)(7)(8)(9)
    - Immunosuppressive agent (eg, tacrolimus, azathioprine, methotrexate, cyclosporin)
    - Immunomodulatory therapy (eg, TNF inhibitor, interleukin inhibitor, monoclonal antibodies)
  - Poorly controlled diabetes mellitus (eg, HbA1c of 8.5% (69 mmol/mol) or higher)[C][D](10)(11)(12)(13)
  - Chronic kidney disease (ie, estimated GFR less than 60 mL/min/1.73m^2 (1.0 mL/sec/1.73m^2)) or end-stage renal disease (hemodialysis or peritoneal dialysis)[E](15)(16)(17)(18)(19)(20)
  - Cirrhosis[F](21)(22)(23)(24)(25)
  - Bone marrow suppression or dysfunction (eg, chemotherapy, radiation therapy, bone marrow fibrosis or infiltration with malignant cells)
  - Humoral (antibody, B-lymphocyte) immune dysfunction or deficiency (eg, common variable immune deficiency, specific antibody deficiency, chronic lymphocytic leukemia)(26)(27)
  - Cell-mediated (eg, T-lymphocyte, natural killer cell) immune dysfunction or deficiency (eg, severe combined immune deficiency, DiGeorge syndrome, Wiskott-Aldrich syndrome)
  - Phagocytic cell (neutrophil, macrophage) dysfunction (eg, chronic granulomatous disease, leukocyte adhesion deficiency)
  - Acquired immunodeficiencies (eg, HIV/AIDS, hematopoietic and lymphoid malignancies)
  - History of solid organ or hematopoietic stem cell transplant[G]
  - Asplenia (surgical or congenital) or functional hyposplenia (eg, sickle cell disease)[H](28)
  - Complement deficiency (eg, hereditary, due to protein loss (nephrotic syndrome))

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Footnotes

A. Examples are given for some of the mechanisms of immunocompromise; however, the list is not all-encompassing. The intent is to illustrate the degree of immunocompromise that may be clinically significant. The presence of a listed potentially immunocompromising factor does not always indicate clinically significant immunosuppression. For example, some findings, such as neutropenia, are usually indicative of significant immunocompromise, whereas others (eg, presence of diabetes, renal failure, or asplenia) may not necessarily indicate clinically important immunocompromise. The judgment as to the presence of clinically significant immunocompromise in a particular patient is also influenced by several variables, such as the nature of the suspected infection and the specific details of the patient's presentation, condition, and comorbidities. Patients may have more than one potential source of immunocompromise, increasing the likelihood of significant dysfunction (eg, poorly controlled diabetes and chronic kidney disease).

B. The minimum dose of systemic corticosteroid use associated with an increased risk of serious infection is not defined, but there is increasing risk with higher doses and longer duration of treatment.(5) Observational studies of patients with rheumatologic disease indicated, after statistical adjustment, that infection risk was increased with prolonged use of as little as 5 mg of prednisone-equivalent per day dosing, and also found that higher doses and longer duration increased risk.(6)(7)(8) An evidence-based specialty society guideline defined clinically important immunocompromise in adult pneumonia patients as those patients with an impaired immune response who are judged to be at elevated risk of infection by atypical or opportunistic organisms, so that there is a need to alter empirical antimicrobial therapy.(9) This same guideline stated that corticosteroid use would lead to such an immune system impairment at doses of 20 mg prednisone-equivalent per day or more for 14 days or more, or a course of treatment with a cumulative dose of 600 mg prednisone-equivalent or more.(9)

C. It is generally agreed that the presence of diabetes is associated with an increased risk of infection, including infection leading to hospitalization.(2)(10)(11)(12)(13) At least some of this risk is attributed to the deleterious effects of diabetes on the immune system.(2)(10)(11)(12)(13) It has been found that the impact of diabetes can be higher in certain types of infections (eg, higher impact in cellulitis, osteomyelitis) and with a longer total duration of the diagnosis.(10)(11)(13)

D. There is no universally agreed upon HbA1c level that indicates "poor control." It has been found that the higher the HbA1c, the larger the impact on the immune system.(10)(11)(12)(13) Goal HbA1c can vary to some degree depending on the patient. Otherwise healthy adults and adolescents may have a target HbA1c of less than 7% (53 mmol/mol), whereas those who have complex comorbidities, impairment in cognition or activities of daily living, or a history of severe hypoglycemia may have a target of less than 8% (64 mmol/mol).(14) However, these targets are put forth with reduction of microvascular and macrovascular effects of diabetes in mind, not immune system effects.(14) In this way, there is no simple cutoff between adequate and poor control. Cohort studies, after statistical adjustment, have found that HbA1c levels above 8.0% to 8.5% (64 to 69 mmol/mol) are independently associated with hospitalization due to infection.(10)(11)(12)(13)

E. Patients with chronic kidney disease or end-stage renal disease (ESRD) have impaired immune system function, measured, for example, by higher rates of infection requiring hospitalization.(15)(16)(17) The degree of renal impairment correlates with the severity of immunocompromise.(15)(16)(17) The degree of proteinuria (eg, urine albumin-to-creatinine ratio) is also independently correlated with the severity of immune system dysfunction. These 2 factors are additive, in that patients with both reduced GFR and significant proteinuria are the most severely affected.(15)(16)(17) Patients with ESRD, in general, have more profound immune system dysfunction, with inadequate dialysis even more detrimental.(18)(19)(20)
F. Patients with cirrhosis of the liver are functionally immunocompromised due to a variety of mechanisms. This immunocompromise correlates with the severity of the cirrhosis and is called cirrhosis-associated immune dysfunction (CAID). Immune dysfunction is also worsened during periods of acute exacerbation of chronic liver disease.(21)(22)(23)(24)(25) CAID is particularly associated with bacterial infections, affecting both the frequency and severity of infection.(23)(24)(25)
G. Susceptibility to infection varies with the immunosuppressive agents used.(4)
H. Patients with asplenia or hyposplenism are at higher risk of infection with encapsulated organisms, such as *Streptococcus pneumoniae*, meningococcal infection, encapsulated strains of *Haemophilus influenzae*, and certain tick-borne diseases.(2)(28)

Orthostatic hypotension

- Orthostatic hypotension,[A][B] as indicated by **1 or more** of the following(1)(2)(3):
  - Fall in SBP of 20 mm Hg or more 1 to 3 minutes after patient sits or stands from recumbent position
  - Fall in DBP of 10 mm Hg or more 1 to 3 minutes after patient sits or stands from recumbent position

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Footnotes

A. Concomitant measurements of the heart rate are important to measure to help diagnose subtypes of orthostatic hypotension (eg, the lack of a compensatory increase in heart rate is typical of autonomic failure and an exaggerated tachycardia may be reflective of volume depletion). However, the heart rate is not a component of the definition of orthostatic hypotension, which relies upon blood pressure alone.(1)(2)(3)
B. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

Respiratory distress

- Respiratory distress, as indicated by **ALL** of the following(1)(2):
  - Patient with **1 or more** of the following:
    - Dyspnea (difficulty breathing)
    - Tachypnea
    - Abnormal breathing pattern (eg, chest retractions)
    - Other evidence of difficulty breathing
  - Evidence of respiratory compromise, as indicated by **1 or more** of the following:
    - Hypoxemia
    - Altered mental status

- Other evidence of respiratory compromise (eg, respiratory acidosis)

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Severe airflow or ventilation abnormalities

- Severe airflow or ventilation abnormalities (not responsive to observation care as appropriate), as indicated by **1 or more** of the following(1)(2)(3)(4):
  - PCO_2 greater than 42 mm Hg (5.6 kPa) and pH less than 7.35 (new)
  - Documented PCO_2 increased more than 5 mm Hg (0.7 kPa) from disease baseline that persists despite appropriate treatment (eg, nebulizer treatments, despite observation care)
  - Airflow measurements[A] less than 60% of previous best or predicted (eg, peak expiratory flow rate less than 300 L/min) that persist despite appropriate treatment (eg, nebulizer treatments, despite observation care)
  - Required respiratory treatments that are performable only in acute inpatient setting

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Footnotes

A. Airflow measurements may include peak expiratory flow rate or forced expiratory volume.(2)(3)

Severe respiratory findings

- Severe respiratory findings (not responsive to observation care as appropriate), including **1 or more** of the following(1)(2)(3):
  - Respiratory distress, as indicated by **ALL** of the following(2)(4):
    - Patient with **1 or more** of the following:
      - Dyspnea (difficulty breathing)
      - Tachypnea
      - Abnormal breathing pattern (eg, chest retractions)
      - Other evidence of difficulty breathing
    - Evidence of respiratory compromise, as indicated by **1 or more** of the following:
      - Hypoxemia
      - Altered mental status
      - Other evidence of respiratory compromise (eg, respiratory acidosis)
  - Gross hemoptysis[A](5)(6)
  - Acute cyanosis

References

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Footnotes

A. Gross or massive hemoptysis is either occurring at a significant rate over time (eg, 200 mL over 24 hours) or is rapid (eg, 100 mL within an hour). Regardless of the amount, which can be difficult to ascertain, clinical factors such as ability of a patient to maintain a patent airway and expectorate blood, the swiftness of available therapeutic options, and the patient's underlying physiologic reserve are all relevant situations to consider in managing hemoptysis.(5)

Tachycardia

- Tachycardia,[A][B] as indicated by **1 or more** of the following(1)(2):
  - Heart rate greater than 100 beats per minute in adult or child age 6 years or older
  - Heart rate greater than 115 beats per minute in child 3 to 5 years of age
  - Heart rate greater than 125 beats per minute in child 1 or 2 years of age
  - Heart rate greater than 130 beats per minute in infant 6 to 11 months of age
  - Heart rate greater than 150 beats per minute in infant 3 to 5 months of age
  - Heart rate greater than 160 beats per minute in infant 1 or 2 months of age

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1. Schriger DL. Approach to the patient with abnormal vital signs. In: Goldman L, Cooney KA, editors. Goldman-Cecil Medicine. 27th ed. Elsevier; 2024:32-35.
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Footnotes

A. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
B. Interpretation of heart rate requires clinical judgment and consideration of several patient-specific factors, such as the patient's baseline heart rate, medications, and clinical impact. For example, an elderly patient on a beta-blocker medication with a baseline resting heart rate of 60 beats per minute may be clinically tachycardic at a heart rate of 94 beats per minute. Likewise, a patient who is upset, in pain, or nervous in the emergency department with a heart rate of 106 beats per minute may meet the technical definition of tachycardia, but this tachycardia (absent associated findings such as chest pain or hypotension) may not be clinically important. The numeric values included in this definition are provided to allow for consistency in terms of a technical definition of the term tachycardia. Whether a heart rate above or below the technical threshold is clinically meaningful is a matter of persistence, context, and clinical judgment.

Tachypnea

- Tachypnea,[A][B] as indicated by respiratory rate of **1 or more** of the following(2)(3):

- Greater than 18 breaths per minute in adult or child 13 years of age or older[A]
- Greater than 22 breaths per minute in child 6 to 12 years of age[A]
- Greater than 25 breaths per minute in child 3 to 5 years of age[A]
- Greater than 30 breaths per minute in child 1 or 2 years of age[A]
- Greater than 40 breaths per minute in infant 6 to 11 months of age[A]
- Greater than 45 breaths per minute in infant 3 to 5 months of age[A]
- Greater than 60 breaths per minute in infant 1 or 2 months of age[A]

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Footnotes

A. Critérios baseados em valores numéricos obtidos pelo médico (por exemplo, sinais vitais, saturação de oxigênio) devem ser utilizados se refletirem com precisão a condição do paciente. Achados transitórios (por exemplo, anormais apenas na admissão inicial no pronto-socorro ou em apenas uma das múltiplas medições) que melhoram rapidamente com pouco ou nenhum tratamento geralmente não refletem a gravidade da doença ou o risco de piora. Isso não significa que uma medição inicial ou isolada nunca possa ser aplicada. O objetivo é diferenciar achados errôneos ou incidentais daqueles que realmente representam o quadro clínico do paciente.

B. A interpretação da frequência respiratória requer julgamento clínico e a consideração de diversos fatores específicos do paciente, como sua frequência respiratória basal e se ela indica alguma patologia respiratória ou outra condição subjacente significativa. Um aumento leve na frequência respiratória (por exemplo, de 20 a 22 em um adulto), na ausência de outros indícios de doença grave (por exemplo, hipoxemia, acidose metabólica, diminuição do volume corrente, dor), pode ser transitório e não ter importância clínica. Os valores numéricos incluídos nesta definição são fornecidos para garantir consistência em termos de uma definição técnica do termo taquipneia. A relevância clínica de uma frequência respiratória acima do limiar técnico depende da persistência, do contexto e do julgamento clínico. Além disso, a medição da frequência respiratória está sujeita a erros. Um livro-texto de medicina afirma que, devido à possibilidade de variação significativa na frequência respiratória entre as respirações, ela deve ser avaliada por pelo menos 30 segundos, preferencialmente por um minuto inteiro.(1) Este mesmo livro didático também afirma que as novas tecnologias desenvolvidas para medir a frequência respiratória não se mostraram clinicamente úteis.(1)

anormalidade nos sinais vitais

- Anormalidade nos sinais vitais, conforme indicado por **TODOS** os seguintes:(1):
  - Resultados dos sinais vitais não condizentes com a condição crônica do paciente ou com seus valores basais (ex.: pressão arterial intencionalmente baixa em insuficiência cardíaca).
  - Anormalidade nos sinais vitais, conforme indicado por **um ou mais** dos seguintes:
    - Taquicardia
    - Hipotensão
    - Hipotensão ortostática

Referências

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